Hardware User Manual

RTEX Slave Module Series

RTEX-PM
RTEX-DI32
RTEX-DO32T
RTEX-AI8
RTEX-AO4

AJINEXTEK AXT

Product Information

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Revision History

Version Info	Date & Version	Comment
Rev. 1.0 issue 1.0	2008.03.21	
Rev. 1.0 issue 1.1	2009.04.20	TB1 pin assignment is changed reversely
Rev. 1.1 issue 1.2	2009.05.26	Specifications and Scheme is modified

1. Overview

1.1. Introduction

RTEX(Real Time Express) Series Controller is high speed distributed network controller made by Panasonic.

RTEX-PM is a module that can control any servo/step drive under RTEX environment, replacing A4N Servo Drive of Panasonic that controls equipments using high speed network communication protocol.

RTEX-DI32 is a slave digital signal module using high speed network communication protocol that includes a function to control digital signal input, and it contains 32 input contacts.

RTEX-DO32T is a slave digital signal module using high speed network communication protocol that includes a function to control digital signal output, and it contains 32 output contacts (transistor output).

RTEX-AI8 is slave analog signal module using high speed network communication protocol that includes a function to control analog signal input, and it is configured to process total eight channels of -10V ~ +10VDC voltage, 0 ~ +5V, 0 ~ +10V (voltage mode) or 4mA ~ 20mA current (current mode) inputs.

RTEX-AO4 is a slave analog signal module using high speed network communication protocol that includes a function to control analog signal output, and it is configured to process total four channels of -10V ~ +10VDC voltage (voltage mode) outputs.

1.2. Application

RTEX Slave Serious modules communicate with RTEX master board using RTEX (Real Time Express) communication protocol and it provides external digital and output signal interfaces and motion control functions, which makes this product ideal for various applications including factory automation, machine tool control, lab experiment, education, etc.

▶ Digital monitoring
▶ Signal conversion
▶ Optical isolation between computer and peripheral devices
▶ Automatic test equipments
▶ Equipment monitoring
▶ Equipment interface
▶ Process (treatment) control
▶ Industrial measurement
▶ Data acquisition from sensor devices
▶ Signal analysis
▶ Industrial digital input / output control
▶ 24 V signal monitoring

2. Specification and Appearance of RTEX Series

2.1. Module Specification and Scheme

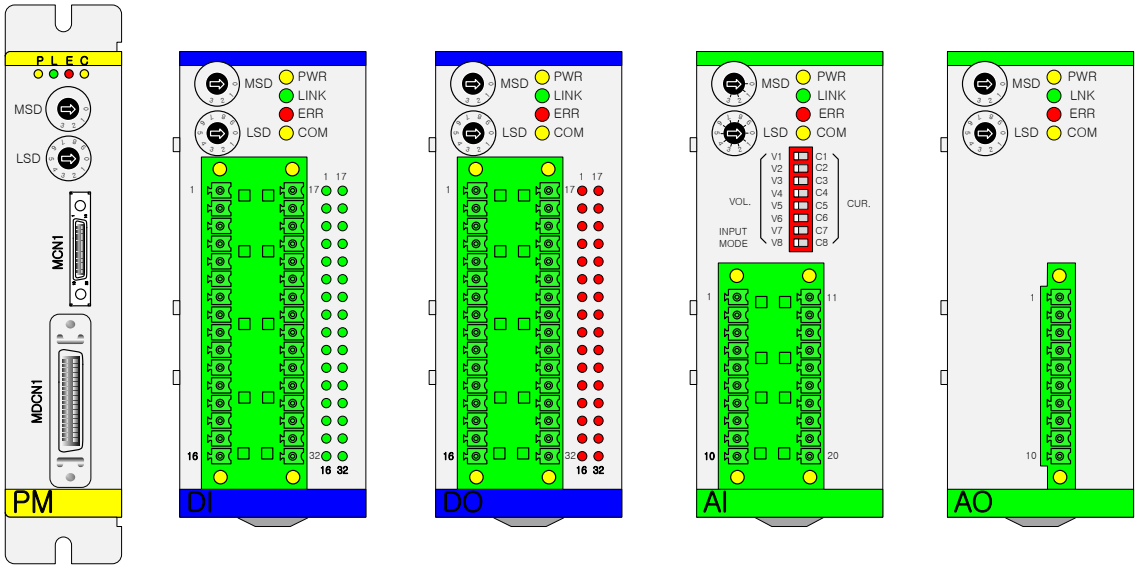


Fig 1. RTEX SLAVE Series Module

Table 1. RTEX-PM Module Specification

Dimension	120mm X 110mm X 25mm (Height X Depth X Width)
Mounting Method	Screw Mounting
Number of Control Axes	One axis
Pulse Output	PULSE+, PULSE-, DIR+, DIR-
	LINE TRANSMITTER(5VDC, Up to 8MPPS)
Motion Input and Output	ALARM, INP, RDY(IN4) : PHOTO-COUPLER(24VDC)
	SVON, ALMCLR, DCC : DALINGTON PHOTO-COUPLER(24VDC)
	LIMIT+, LIMIT-, ORG : PHOTO-COUPLER(24VDC)
	TRG+, TRG- : LINE TRANSMITTER(5VDC)
	PCOM, NCOM : DIGITAL I/O Common Input (24VDC LEVEL)
Universal Input and Output	IN2, IN3 : PHOTO-COUPLER(24VDC)
	OUT2, OUT3 : PHOTO-COUPLER + DRIVE IC(24VDC, Up to 50mA/Contact)
Encoder Input	A, /A, B, /B, Z, /Z : High Speed PHOTO COUPLER (5VDC, Up to 8MPPS)
	ENCPWR, LGND : 5VDC Output (when using encoder for step motor)
Node ID Change	Decimal rotary switch (MSD, LSD), (0~31)

LED Display	PWR(Power, Yellow), LINK(Link, Green), ERR(Error, Red), COM(Communication, Yellow)
Connector	Motion: HONDA 26PIN X 1 EA, Motion IO : 3M 26PIN X 1 EA
Power Supply for External Input / Output	PCOM, EMGSTOP : 24VDC LEVEL
	+24V, 24VG : Internal logic power supply (Isolated DC/DC Converter)
	EMGSTOP (Emergency stop) : PHOTO-COUPLER (24VDC) (Used for motion drive)

Table 2. RTEX-DI32 Module Specification

Dimension	100mm X 90mm X 40mm (Height x Depth x Width)
Mounting Method	DIN Rail
DIGITAL Input	Total 32 input contacts
	Input: PHOTO-COUPLER (24VDC)
	Connector: Phoenix 16 Pin x 2 EA
LED Display	PWR (Power, Yellow), LINK (Link, Green), ERR (Error,Red), COM (Communication, Yellow)
	IN1 ~ IN32(Green)
Node ID Change	Decimal rotary switch (MSD, LSD), (0~31)
Power Supply for External Input and Output	PCOM, NCOM : DIGITAL I/O Common Input (24VDC LEVEL)
	+24V, 24VG : Internal Logic Power Supply (Non-isolated Switching Power Supply)

Table 3. RTEX-DO32 Module Specification

Dimension	100mm X 90mm X 40mm (Height x Depth x Width)
Mounting Method	DIN Rail
DIGITAL Output	Total 32 output contacts
	Output: PHOTO-COUPLER + DRIVE IC(24VDC, Up to 50mA/contacts)
	Connector: Phoenix 16 Pin x 2 EA
LED Display	PWR (Power, Yellow), LINK (Link, Green), ERR (Error, Red), COM (Communication, Yellow)
	OUT1 ~ OUT32(Red)
Node ID Change	Decimal rotary switch (MSD, LSD), (0~31)
Power Supply for	PCOM, NCOM : DIGITAL I/O Common Input (24VDC LEVEL)

External Input and Output	+24V, 24VG : Internal Logic Power Supply(Non-isolated Switching Power Supply)

Table 4. RTEX-AI8 Module Specification (To be released)

Dimension	100mm X 90mm X 40mm (Height x Depth x Width)
Mounting Method	DIN Rail
Number of inputs	8 channels
	SINGLE-ENDED INTERFACE CIRCUIT
Resolution	16BIT
Sampling speed	10KHz
Input range	Voltage mode: One of -10V ~ +10VDC, 0~5V or 0~10V can be selected.
	Current mode : 4mA ~ 20mA (when the front panel current mode switch is turned on).
	Connector: Phoenix 10 Pin x 2 EA
LED Display	PWR (Power, Yellow), LINK (Link, Green), ERR (Error, Red), COM (Communication, Yellow)
Node ID Change	Decimal rotary switch (MSD, LSD), (0~31)
Power Supply for External Input and Output	+24V, 24VG : Internal Logic Power Supply(Isolated DC/DC Converter)

Table 5. RTEX-AO4 Module Specification(To be released)

Dimension	100mm X 90mm X 40mm (Height x Depth x Width)
Mounting Method	DIN Rail
Number of inputs	4 channels
	Voltage mode
Resolution	16BIT
Sampling speed	10KHz
Input range	Voltage mode : -10V ~ +10V
	Connector: Phoenix 10 Pin x 1 EA
LED Display	PWR (Power, Yellow), LINK (Link, Green), ERR (Error, Red), COM (Communication,Y ellow)
Node ID Change	Decimal rotary switch (MSD, LSD), (0~31)

Power Supply for External Input and Output	+24V, 24VG : Internal Logic Power Supply(Isolated DC/DC Converter)

2.2. Dimension and Appearance of RTEX Slave Series Modules

2.2.1. RTEX-PM Module

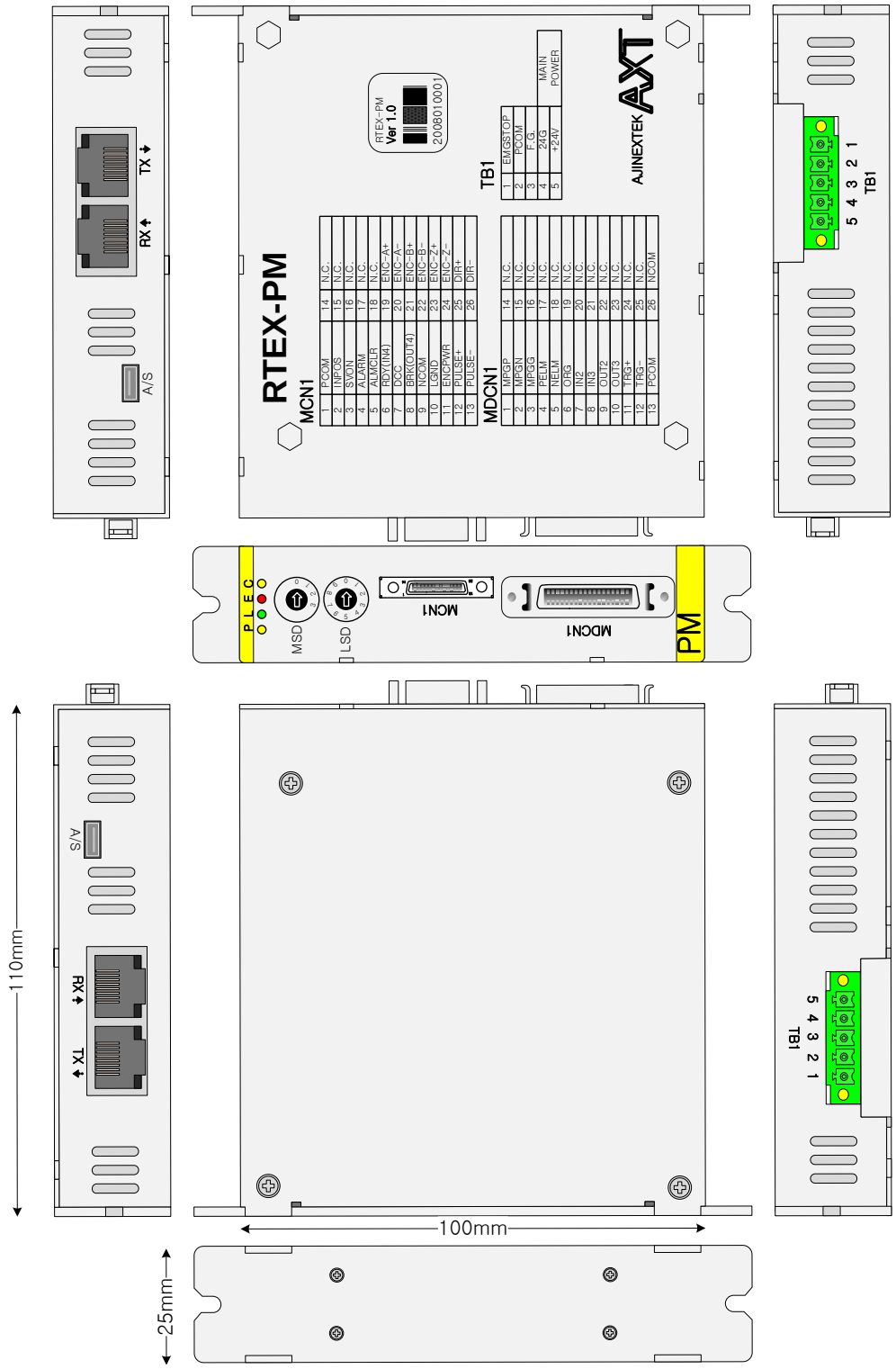


Fig. 2. Dimension and Appearance of RTEX-PM Module

2.2.2. RTEX-DI32 Module

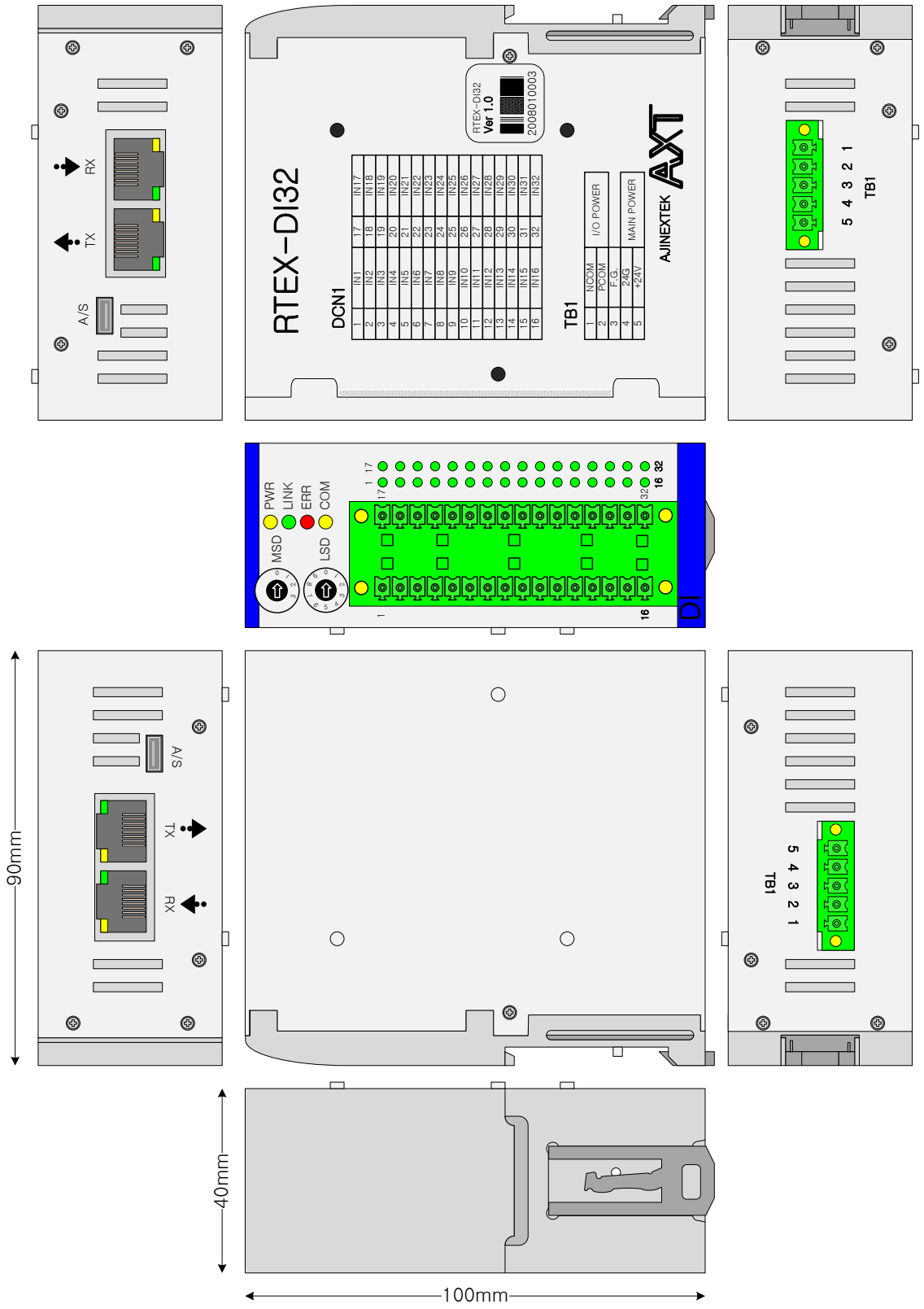


Fig. 3. Dimension and Appearance of RTEX-DI32 Module

2.2.3. RTEX-DO32T Module

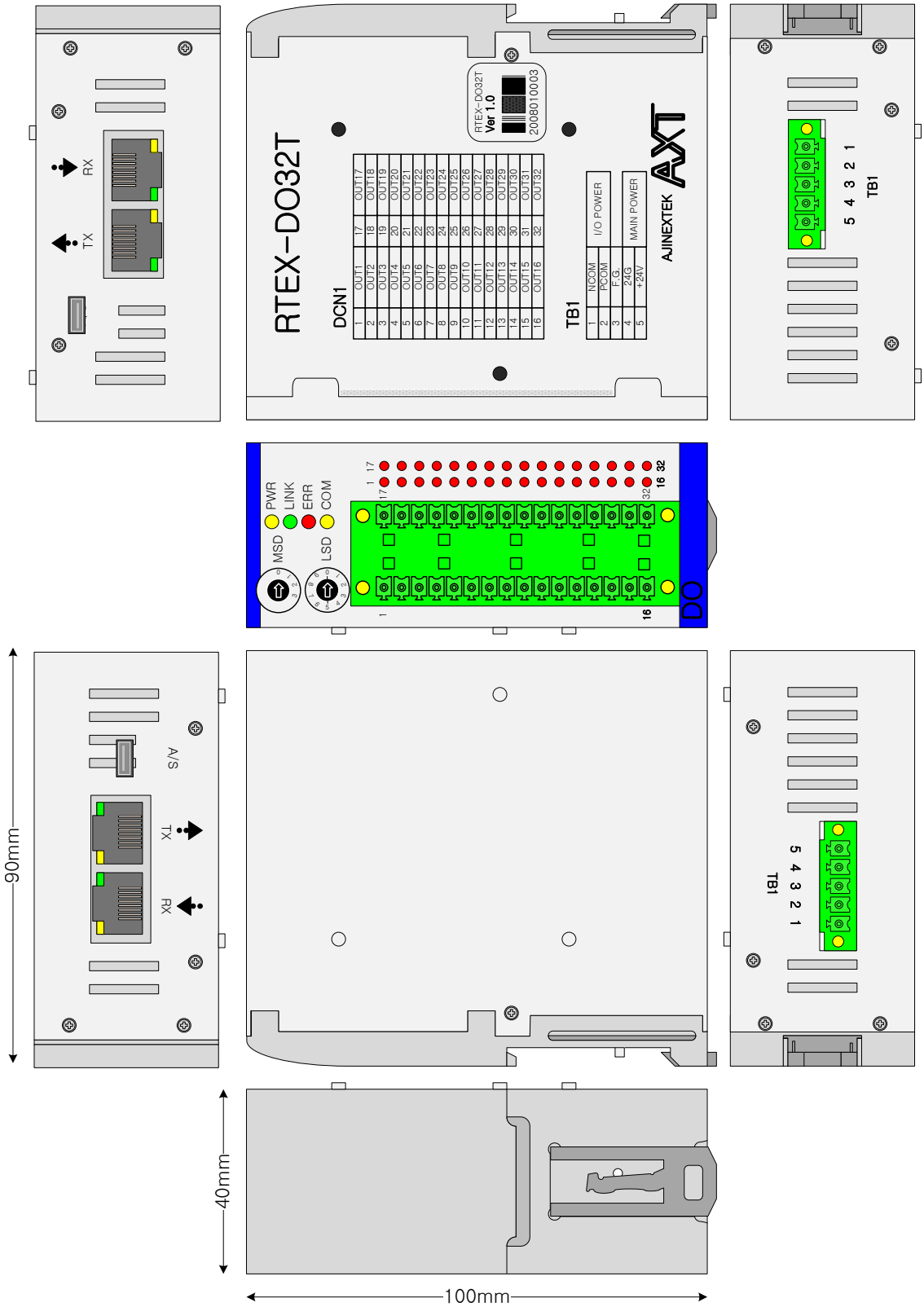


Fig. 4. Dimension and Appearance of RTEX-DO32T Module

2.2.4. RTEX-AI8 Module (To be released)

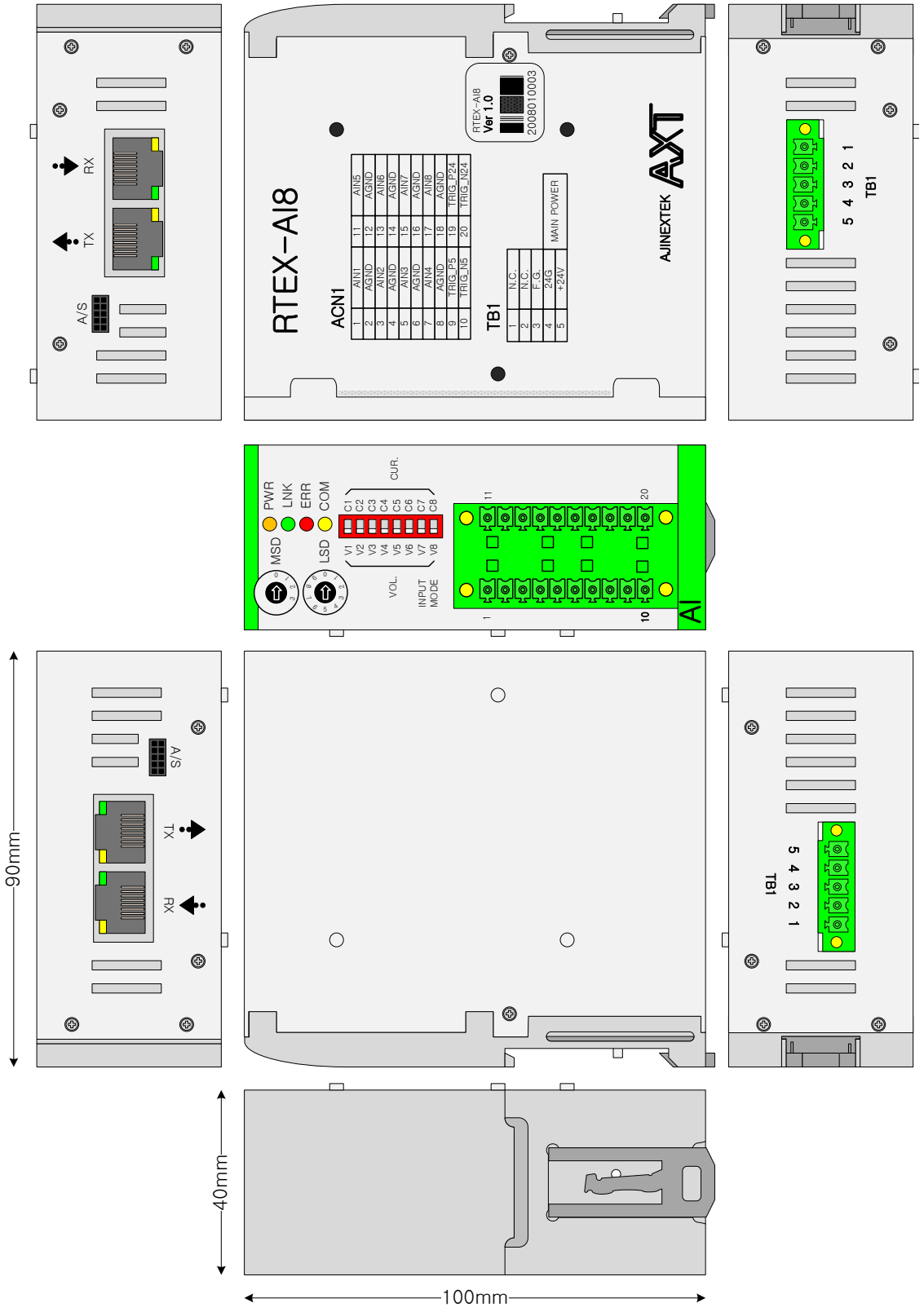


Fig. 5. Dimension and Appearance of RTEX-AI8 Module (To be released)

2.2.5. RTEX-AO4 Module (To be released)

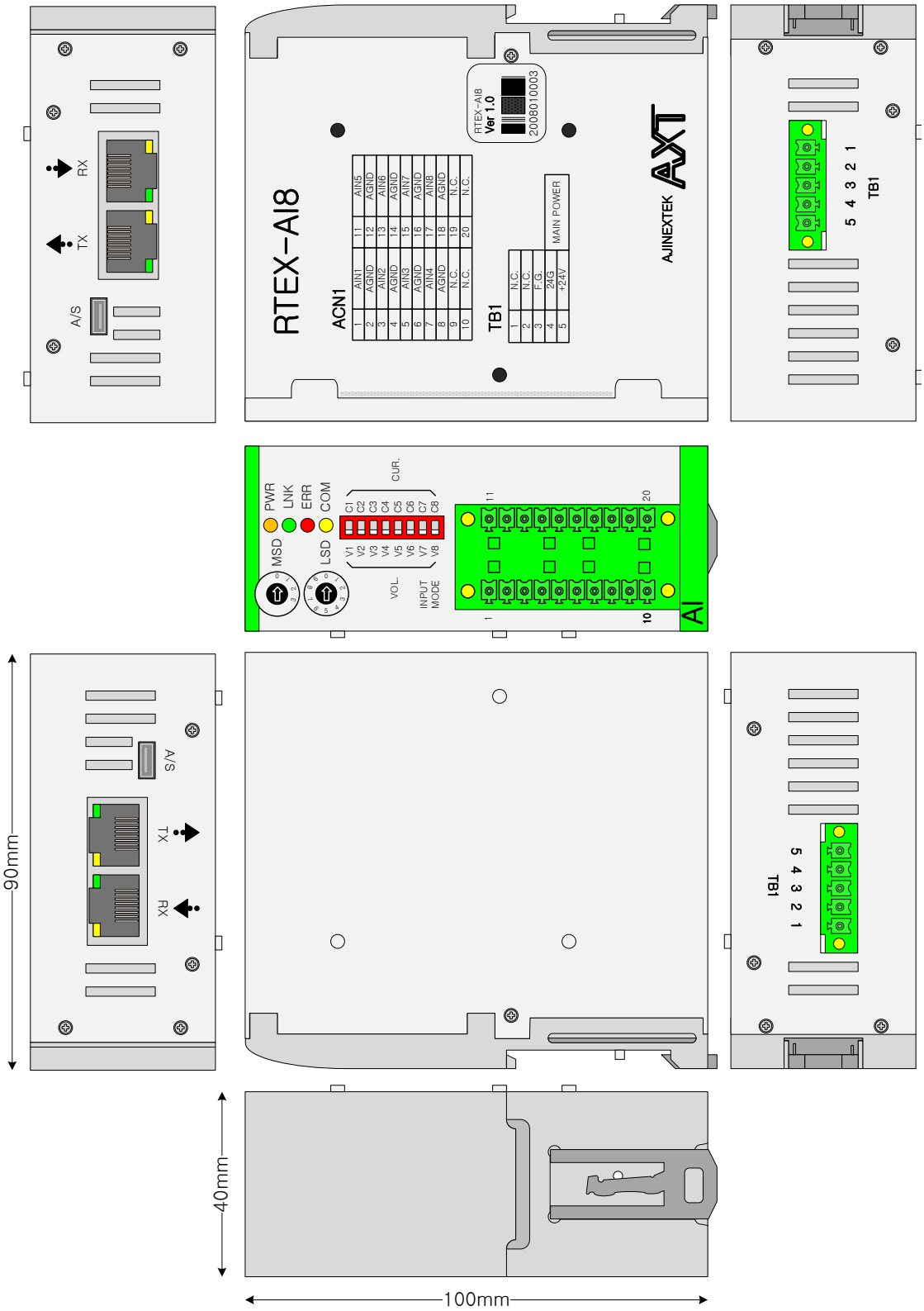


Fig. 6. Dimension and Appearance of RTEX-AO4 Module (To be released)

3. Installation

3.1. Safety Precautions

Make sure to be fully familiar with the following instructions for the efficient and safe use of the product before the use.

3.1.1. Warning

- Ignoring associated instructions would cause serious situation that can lead to death or serious injury.
- Do not assemble the product while the power is applied. Otherwise, it could cause fire, electric shock or malfunction.
- Do not work on wiring while the power is applied. Otherwise, it could cause electric shock or malfunction.
- Do not let metallic foreign substances into the product. Otherwise, it could cause fire, electric shock or malfunction.
- Contact professional electricians for wiring works. Improper handling could cause fire, electric shock or malfunction.

3.1.2. Cautions

- Ignoring associated instructions could cause minor injuries, product damages or property damages.
- Check the voltage and current ratings of the product before wiring. Otherwise, it could cause fire, electric shock or malfunction.
- Avoid the place with ambient temperature over 55℃. Otherwise, it could cause fire, electric shock or malfunction.
- Avoid the place that is exposed to direct sunlight. Otherwise, it could cause fire, electric shock or malfunction.
- Avoid the place with ambient humidity over 85%. Otherwise, it could cause fire, electric shock or malfunction.
- Avoid the place near to flammable substance. Otherwise, it could cause fire.
- Install the product where the vibration is not directly applied to the product. Otherwise, it could cause fire or electric shock.
- Unauthorized disassembly, repair or modification of the product should be prohibited. Otherwise, it could cause fire or electric shock.
- Use the product only under the environment conditions specified in the user's manual. Otherwise, it could cause electric shock, fire, malfunction or product degradation.
- Connect the output load not exceeding the specified rating. Otherwise, it could cause fire, malfunction or trouble.

3.2. Installation Procedure

3.2.1. DIN Rail Mounting and Detachment Method of the Module

RTEX-DI32, RTEX-DO32T, RTEX-AI8 and RTEX-AO4 Modules are designed to be mounted on DIN Rail. (Refer to Fig. 8) Care should be taken for RTEX-PM, pulse type motion slave module, since it is a fixed type and it is not designed to be mounted on DIN Rail.

Follow the procedures described below in installing the product and in reverse order when detaching the product.

- **Installation (Refer to Fig. 8)**

1) Firmly fix the DIN Rail at the desired position.
2) Pull down the rail fixing pin located in the back of the module.
3) Tilt the product slightly while fixing pin is pulled down.
4) Insert the module to DIN Rail from the top of the module.
5) Check whether the top of the module is securely inserted to the DIN Rail and gently attach the bottom part to DIN Rail.
6) Check whether both top and bottom parts of the module are attached completely to the DIN Rail and push up the fixing pin.

- **Detachment (Refer to Fig. 9)**

1) Pull down the fixing pin located at the bottom of the installed module using a proper tool.
2) Tilt the bottom of the module forward and pull it gently.
3) Lift up the module gently so that top of the module is detached from DIN Rail.
4) Push up the module fixing pin.

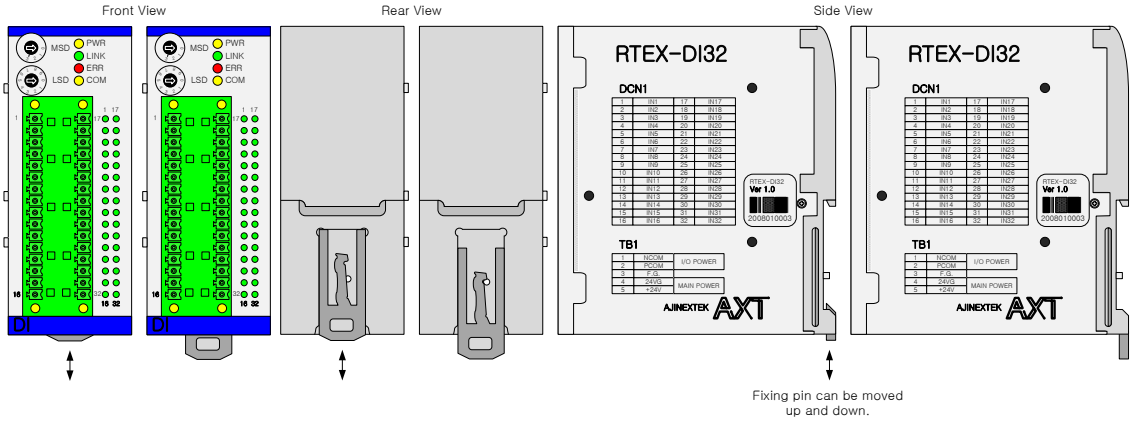


Fig. 7. Module fixing pin

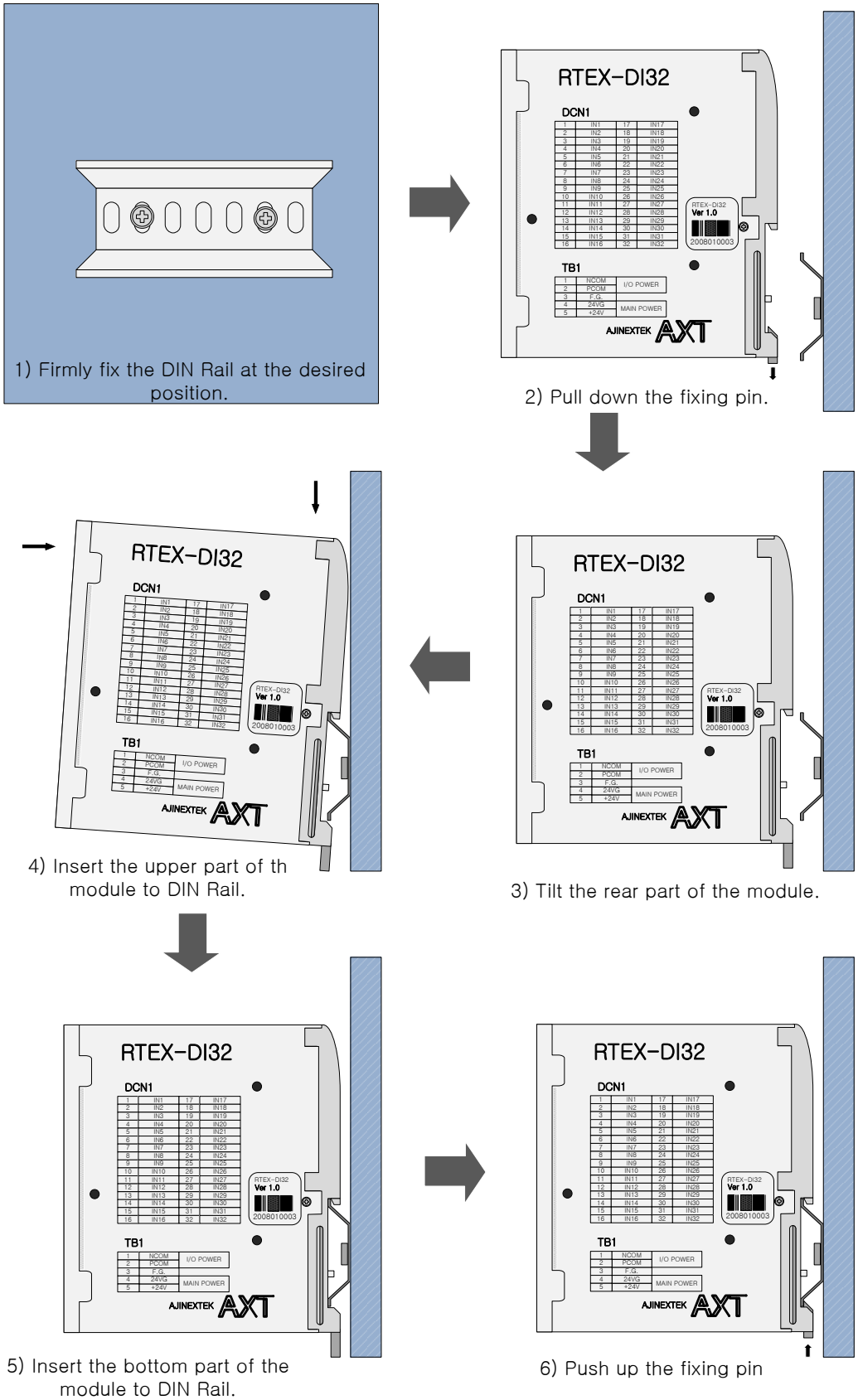


Fig. 8. Module Installation Procedure

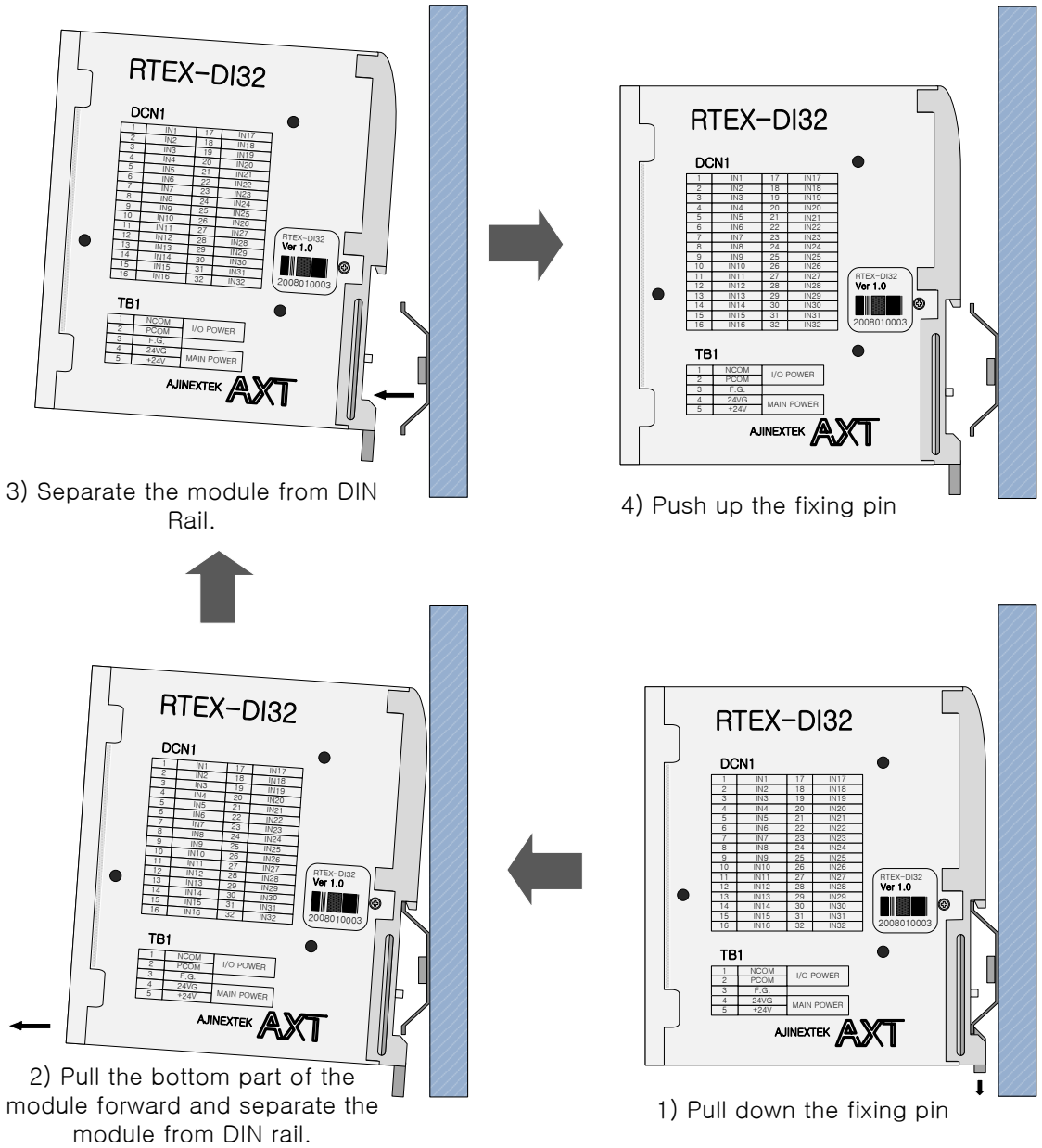
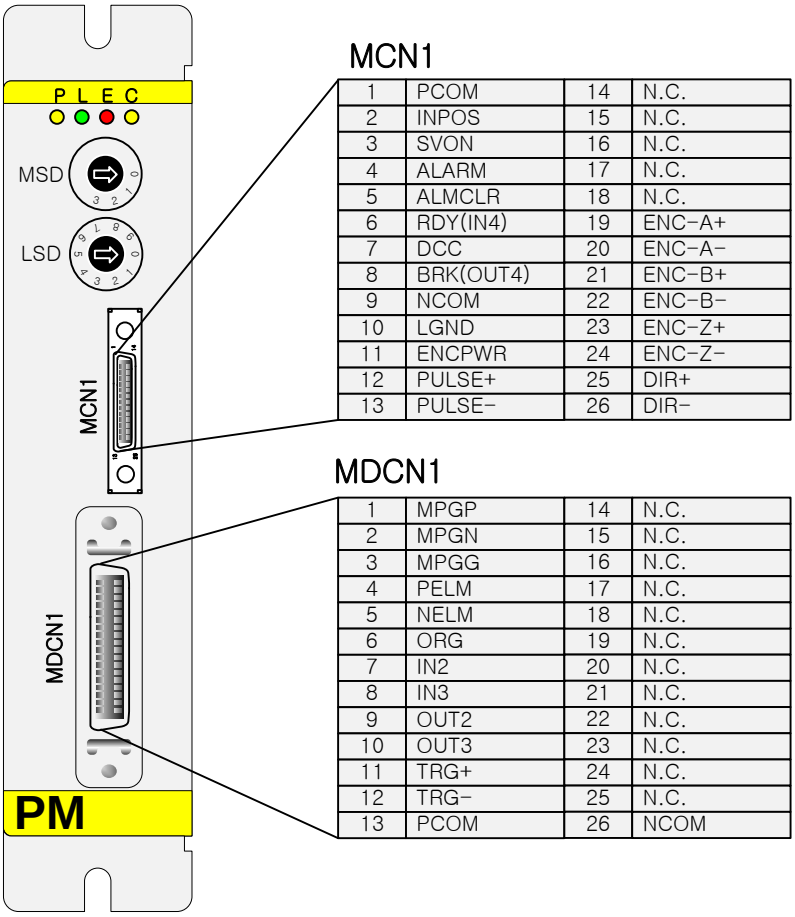


Fig. 9. Module Detachment Procedure

4. Connector Pin Arrays and Descriptions for RTEX Series Modules

4.1. RTEX-PM Module

4.1.1. Pin Array and Description of Front Panel Connector (MCN1, MDCN1)



MCN1

1	PCOM	14	N.C.
2	INPOS	15	N.C.
3	SVON	16	N.C.
4	ALARM	17	N.C.
5	ALMCLR	18	N.C.
6	RDY(IN4)	19	ENC-A+
7	DCC	20	ENC-A-
8	BRK(OUT4)	21	ENC-B+
9	NCOM	22	ENC-B-
10	LGND	23	ENC-Z+
11	ENCPWR	24	ENC-Z-
12	PULSE+	25	DIR+
13	PULSE-	26	DIR-

MDCN1

1	MPGP	14	N.C.
2	MPGN	15	N.C.
3	MPGG	16	N.C.
4	PELM	17	N.C.
5	NELM	18	N.C.
6	ORG	19	N.C.
7	IN2	20	N.C.
8	IN3	21	N.C.
9	OUT2	22	N.C.
10	OUT3	23	N.C.
11	TRG+	24	N.C.
12	TRG-	25	N.C.
13	PCOM	26	NCOM

Fig. 10. Pin Array of RTEX-PM Module Front Panel Connector(MDCN1, MCN1)

As shown in Fig. 10, signals for pulse type control of servo or step motor (position control) for one axis are arranged by axis on MCN1 connector in the front panel of RTEX-PM Module. Other digital input/output signals are arranged on MDCN1 connector. The 26 pin HDR-EC26LFDT2-SLG+ connector of Honda is used for MCN1 with the pin numbering staring from top left as pin 1, bottom left as pin 13, top right as pin 14 and bottom right as pin 26. Refer to **"5.1. Major Functions of RTEX-PM Module"** for more detailed internal circuit.

4.1.2. Description of Front Panel Node ID

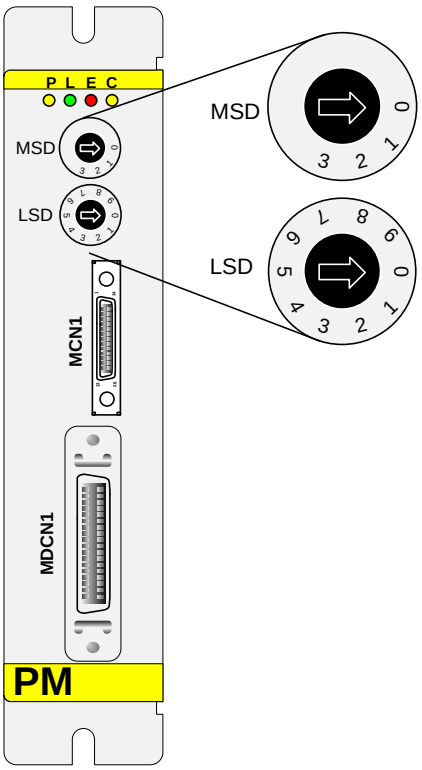


Fig. 11. Front Panel Node ID Switch of RTEX-PM Module

As shown in Fig 11, Node ID switch located in the front panel of the RTEX-PM Module is for setting slave node ID for RTEX communication. Each RTEX slave module should have its own ID before connecting to the master board and the number should be unique among other modules. Node number can be set to any number between 0 and 31. (Connection will fail if the ID is out of this range.) Set the corresponding rotary switch using a proper tool. The following example is to set the module ID to 27.

Setting Example) Setting Node ID to 27: Set the rotary switch of MSD to 2 and LSD to 7. (Refer to Fig 12)

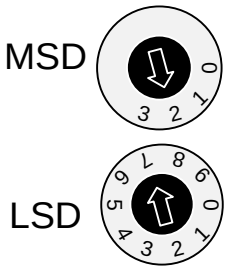


Fig. 12. Setting Node ID of RTEX-PM Module

4.1.3. Pin Array and Description of Top Panel Connector (TX, RX)

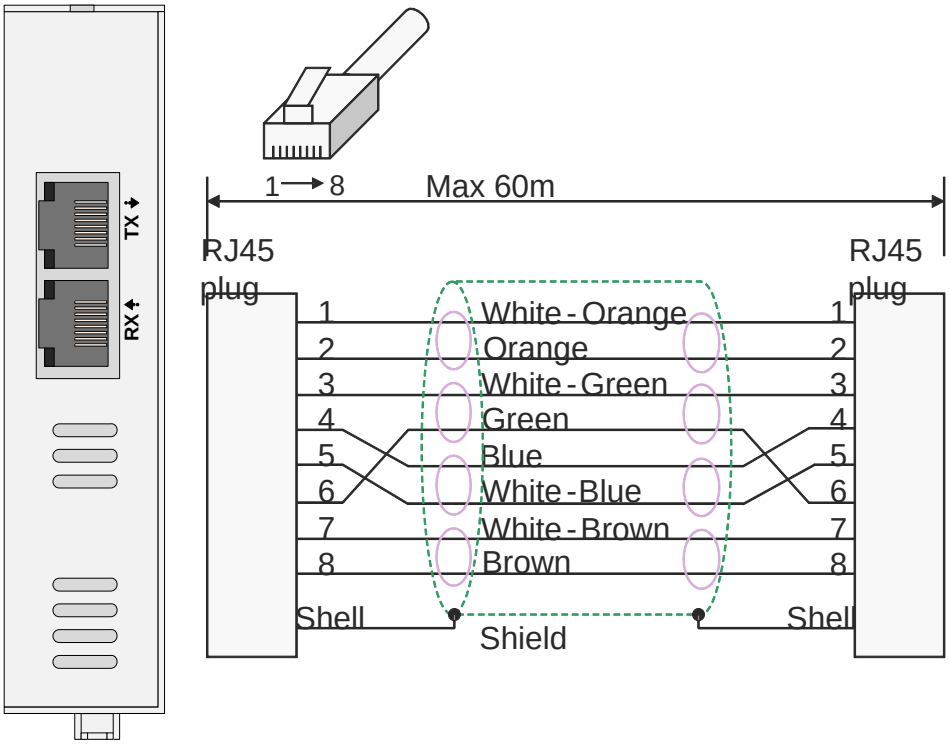


Fig. 13. RJ45 RTEX Communication Port (TX, RX) or RTEX-PM Module

There are two RJ45 ports (TX and RX) for RTEX communication at the top panel of RTEX-PM as shown in Fig 13. Take caution when connecting the master or other slaves. Any direct cable available on the market can be used for communication. However, STP cable is recommended for reliable connection and stability against external noise.

4.1.4. Pin Array and Description of Bottom Panel Connector (TB1)

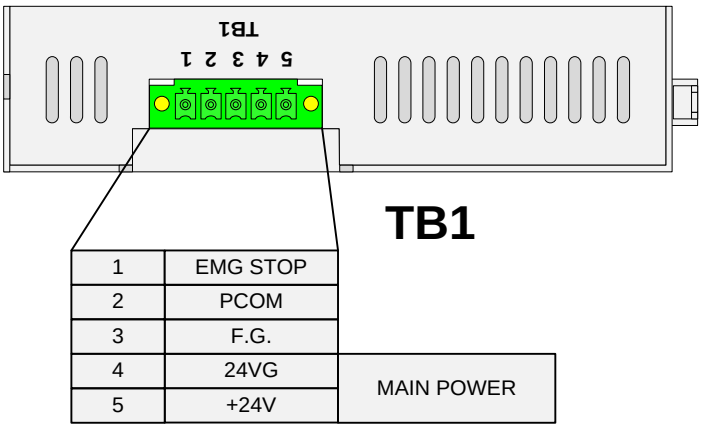
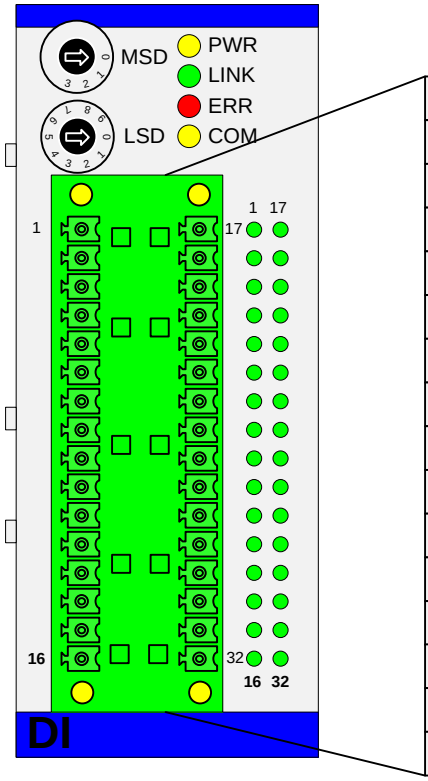


Fig. 14. Pin Array of Bottom Panel Power Connector (TB1) of RTEX-PM Module

There is a power input connector (TB1) at the bottom panel of RTEX-PM as shown in Fig. 14. One MC 1,3/5-GF-3,5THT connector of Phoenix is used for power input connector. Input power is 24V DC. Reverse polarity protection diode is installed inside the module so that power is not supplied to the module and the module will not function if polarity is reversed. Pin 1 and 2 are for emergency stop during motion control and pin 4 and 5 are for external power.

4.2. RTEX-DI32 Module

4.2.1. Pin Array and Description of Front Panel Connector(DCN1)



DCN1

1	IN1	17	IN17
2	IN2	18	IN18
3	IN3	19	IN19
4	IN4	20	IN20
5	IN5	21	IN21
6	IN6	22	IN22
7	IN7	23	IN23
8	IN8	24	IN24
9	IN9	25	IN25
10	IN10	26	IN26
11	IN11	27	IN27
12	IN12	28	IN28
13	IN13	29	IN29
14	IN14	30	IN30
15	IN15	31	IN31
16	IN16	32	IN32

Fig. 15. Pin Array of RTEX-DI32 Front Panel Connector(DCN1)

DCN1 connector located at the front panel of the RTEX-DI32 Module has 32 signals for 32 input contacts as shown in Fig 15. The 32 pin MCD 1,5/16-GIF-3,81 connector of Phoenix is used for DCN1 with the pin numbering staring from top left as pin 1, bottom left as pin 16, top right as pin 17 and bottom right as pin 32. Refer to **"5.2.1. Configuration and Description of Digital Input Circuit of RTEX-DI32 Module"** for detailed internal circuit configuration.

4.2.2. Pin Array and Description of Front Panel LED Display

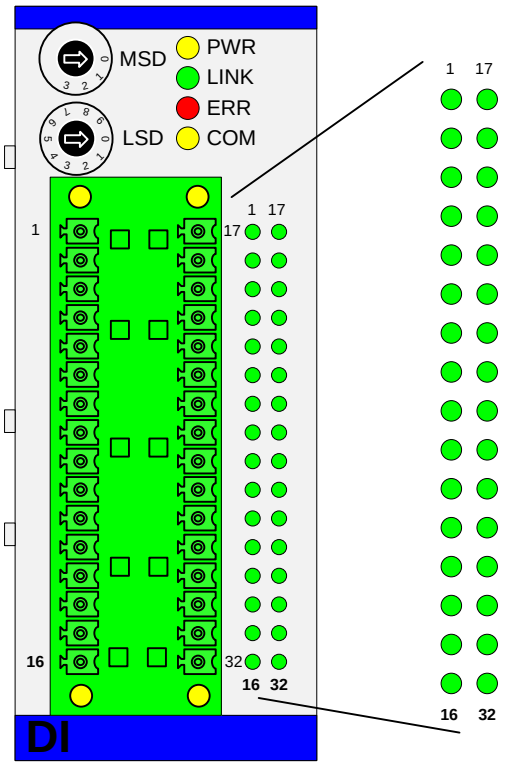


Fig. 16. RTEX-DI32 Front Panel LED Display

As shown in Fig. 16, there are four LED at the upper part and 32 LED on the right side of the front panel of RTEX-DI32 Module. Four LED at the upper part display the status of the slave module and the meaning of each light is as followings.

1) PWR (POWER) : It is on when the power is supplied to the module.
2) LINK (LINK) : It is on when the network is connected.
3) ERR (ERROR) : It displays error status.
4) COM (COMMUNICATION) : It displays the status of RTEX data communication.

LED on the right side display the status of digital input.

1) It is on when input level becomes “L” level and off for “H” level.

Refer to **“5.2.1. Configuration and Description of Digital Input Circuit of RTEX-DI32 Module”** for detailed internal circuit.

4.2.3. Description of Front Panel MODULE ID

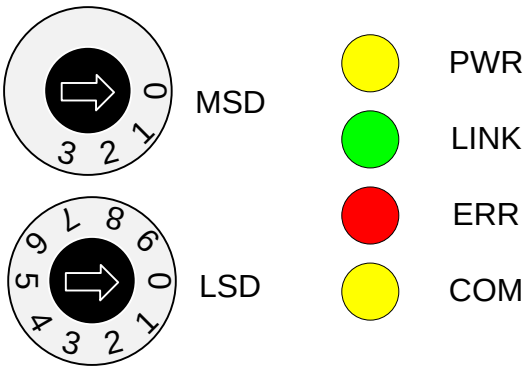


Fig. 17. Module ID Switch of RTEX-DI32 Module

As shown in Fig. 17, there are switches to set the Module ID at the top panel of the RTEX-DI32 Module. Up to 32 slave modules can be connected to one master board and the module ID should be set to 0 to 31. One of the most important things in setting node ID is to set a unique node ID that is not overlapped with other function modules. If there are duplicated ID within a node, it causes an error and entire modules may not work or system may not function properly.

4.2.4. Pin Array and Description of Bottom Panel Connector (TB1)

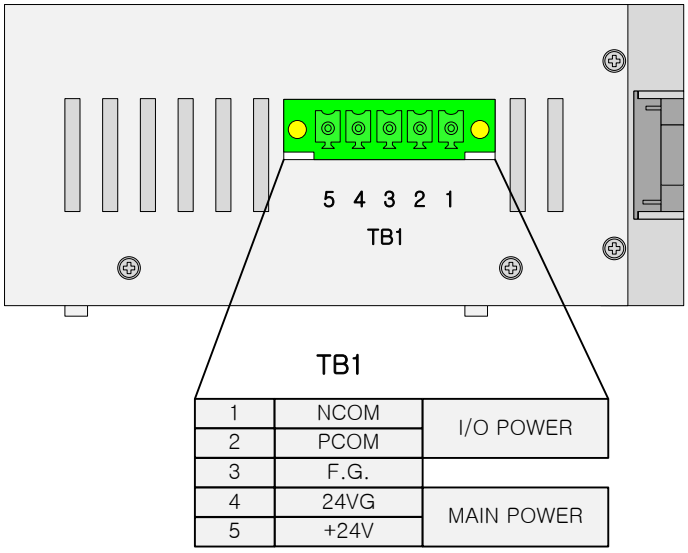
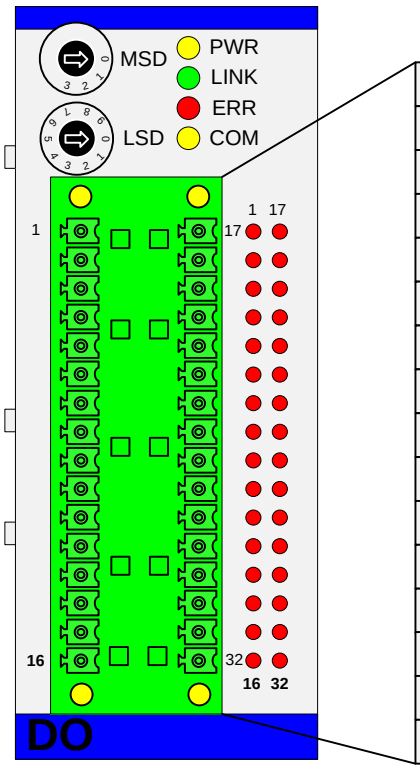


Fig. 18. Pin Array of Bottom Panel Power Connector of RTEX-DI32 Module (TB1)

As shown in Fig. 18, there is a power input connector (TB1) at the bottom panel of RTEX-DI32. One five pin MC 1,3/5-GF-3,5THT connector of Phoenix is used for power input connector. Input power is 24V DC. Reverse polarity protection diode is installed inside the module so that power is not supplied to the module and the module will not function if polarity is reversed. Pin 1 and 2 are for IO power and pin 4 and 5 are for external power.

4.3. RTEX-DO32T Module

4.3.1. Pin Array and Description of Front Panel Connector (DCN1)



DCN1

1	OUT1	17	OUT17
2	OUT2	18	OUT18
3	OUT3	19	OUT19
4	OUT4	20	OUT20
5	OUT5	21	OUT21
6	OUT6	22	OUT22
7	OUT7	23	OUT23
8	OUT8	24	OUT24
9	OUT9	25	OUT25
10	OUT10	26	OUT26
11	OUT11	27	OUT27
12	OUT12	28	OUT28
13	OUT13	29	OUT29
14	OUT14	30	OUT30
15	OUT15	31	OUT31
16	OUT16	32	OUT32

Fig. 19. RTEX-DO32T Front Panel Connector (DCN1) Pin Array

DCN1 connector located at the front panel of the RTEX- DO32T Module has 32 signals for 32 output contacts as shown in Fig 19. The 32 pin MCD 1,5/16-GIF-3,81 connector of Phoenix is used for DCN1 with the pin numbering staring from top left as pin 1, bottom left as pin 16, top right as pin 17 and bottom right as pin 32. Refer to **"5.3.1. Configuration and Description of Digital Output Circuit of RTEX-DI32 Module"** for detailed internal circuit configuration.

4.3.2. Pin Array and Description of Front Panel LED Display

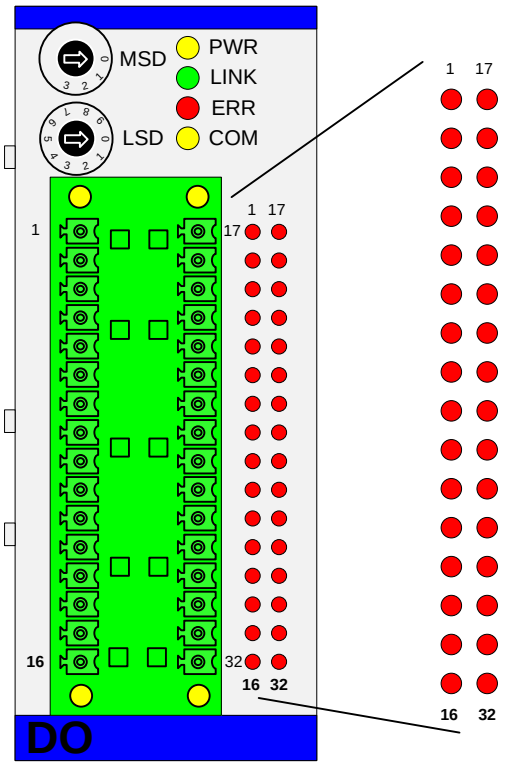


Fig. 20. RTEX-DO32T Front Panel LED Display

As shown in Fig. 20, there are four LED at the upper part and 32 LED on the right side of the front panel of RTEX-DO32T Module. Four LED at the upper part display the status of the slave module and the meaning of each light is as followings.

1) PWR (POWER) : It is on when the power is supplied to the module.
2) LINK (LINK) : It is on when the network is connected.
3) ERR (ERROR) : It displays error status.
4) COM (COMMUNICATION) : It displays the status of RTEX data communication.

LED on the right side display the status of digital output.

1) It is on when output level becomes “L” level and off for “H” level.

Refer to **“5.3.1. Configuration and Description of Digital Output Circuit of RTEX-DI32 Module”** for detailed internal circuit.

4.3.3. Description of Top Panel MODULE ID

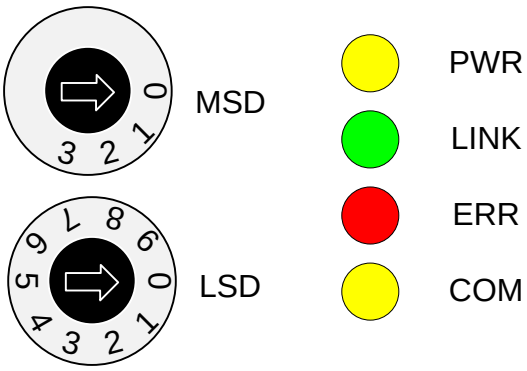


Fig. 21. MODULE ID Switch of RTEX-DO32T

As shown in Fig. 21, there are switches to set the Module ID at the top panel of the RTEX-DO32T Module. Up to 32 slave modules can be connected to one master board and the module ID should be set to 0 to 31. One of the most important things in setting node ID is to set a unique node ID that is not overlapped with other function modules. If there are duplicated ID within a node, it causes an error and entire modules may not work or system may not function properly.

4.3.4. Pin Array and Description of Bottom Panel Connector (TB1)

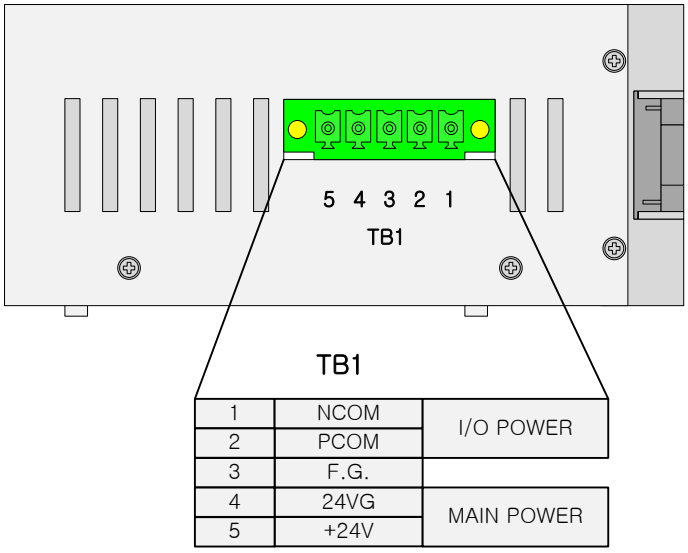
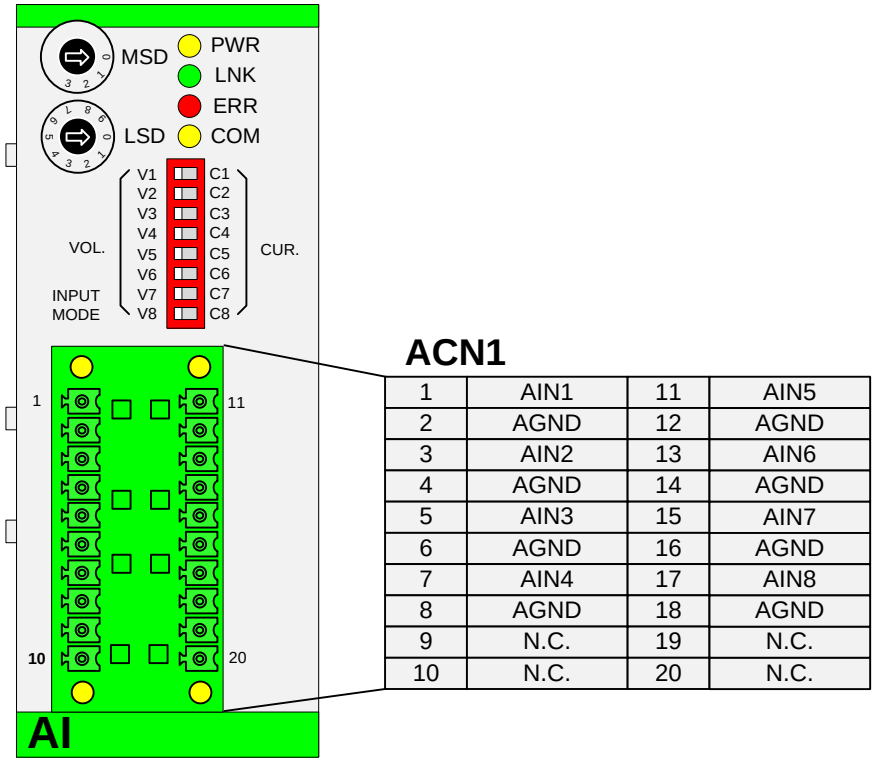


Fig. 22. Pin Array of Bottom Power Connector of RTEX-DO32T Module (TB1)

As shown in Fig. 22, there is a power input connector (TB1) at the bottom panel of RTEX-DO32T. One five pin MC 1,3/5-GF-3,5THT connector of Phoenix is used for power input connector. Input power is 24V DC. Reverse polarity protection diode is installed inside the module so that power is not supplied to the module and the module will not function if polarity is reversed. Pin 1 and 2 are for IO power and pin 4 and 5 are for external power.

4.4. RTEX-AI8 Module (To be released)

4.4.1. Pin Array and Description of Front Panel Connector (ACN1)



ACN1

1	AIN1	11	AIN5
2	AGND	12	AGND
3	AIN2	13	AIN6
4	AGND	14	AGND
5	AIN3	15	AIN7
6	AGND	16	AGND
7	AIN4	17	AIN8
8	AGND	18	AGND
9	N.C.	19	N.C.
10	N.C.	20	N.C.

Fig. 23. Pin Array of RTEX-AI8 Front Panel Connector (ACN1)

As shown in Fig. 23, RTEX-AI8 is an analog input slave module that receives analog voltage signal and the input pins are arranged in such a way that pin 1 and pin 2 are for channel 1, pin 3 and pin 4 are for channel 2, pin 5 and pin 6 are for channel 3, pin 7 and pin 8 are for channel 4, pin 11 and pin 12 are for channel 5, pin 13 and pin 14 are for channel 6, pin 15 and pin 16 are for channel 7, and pin 17 and pin 18 are for channel 8. One 20 pin MCD 1,5/20-GIF-3,81 connector of Phoenix is used for ACN1 connector with the pin numbering staring from top left as pin 1, bottom left as pin 10, top right as pin 11 and bottom right as pin 20. Refer to **"5.4. Major Functions of RTEX-AI8 Module"** for detailed internal circuit.

4.4.2. Pin Array and Description of Front Panel LED Display

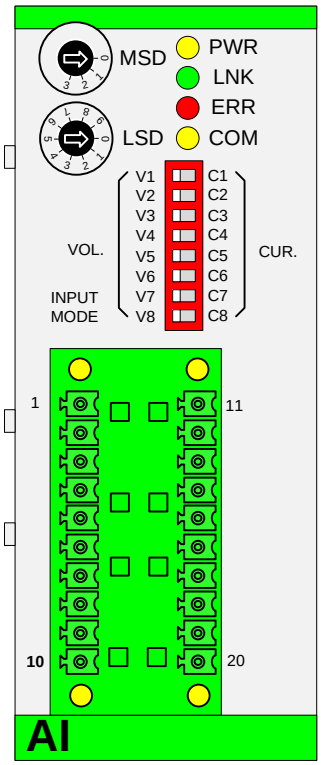


Fig. 24. RTEX-AI8 Front Panel LED Display

As shown in Fig. 24, there are four LED at the upper part and 32 LED on the right side of the front panel of RTEX-AI8 Module. Four LED at the upper part display the status of the slave module and the meaning of each light is as followings.

1) PWR (POWER) : It is on when the power is supplied to the module.
2) LINK (LINK) : It is on when the network is connected.
3) ERR (ERROR) : It displays error status.
4) COM (COMMUNICATION) : It displays the status of RTEX data communication.

4.4.3. Description of Input Mode Selection Switch in the Front Panel

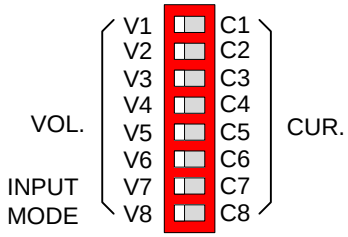


Fig. 25. Input Model Selection Switch of RTEX-AI8 Module

As shown in Fig. 25, there are total eight switches to set the analog input mode of each channel in the front panel of

the RTEX-AI8 Module. Set the switches to V1 ~ V8 for voltage mode and C1 ~ C8 for current mode. That is, set the switch to the left if the analog signal to measure is voltage mode such as -10V ~ +10V, 0 ~ +5V or 0 ~ +10V, or set the switch to the right if it is 4~20mA current. The setting has to be correct depending on the type of signal to measure, otherwise the module may not work correctly.

Refer to **"5.4.1. Configuration and Description of Analog Input Circuit of RTEX-AI8 Module"** for detailed internal circuit.

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4.4.4. Description of Front Panel Module ID

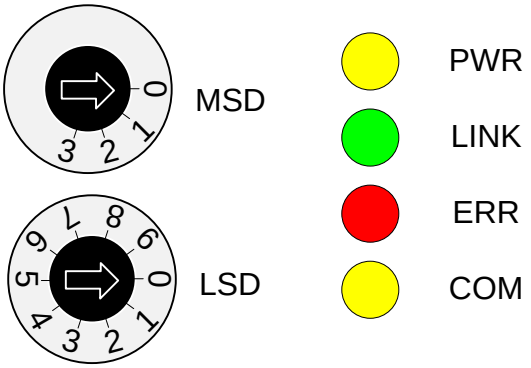


Fig. 26. Module ID Switch of RTEX-AI8 Module

As shown in Fig. 26, there are switches to set the Module ID at the top panel of the RTEX- AI8 Module. Up to 32 slave modules can be connected to one master board and the module ID should be set to 0 to 31. One of the most important things in setting node ID is to set a unique node ID that is not overlapped with other function modules. If there are duplicated ID within a node, it causes an error and entire modules may not work or system may not function properly.

4.4.5. Pin Array and Description of Bottom Panel Connector (TB1)

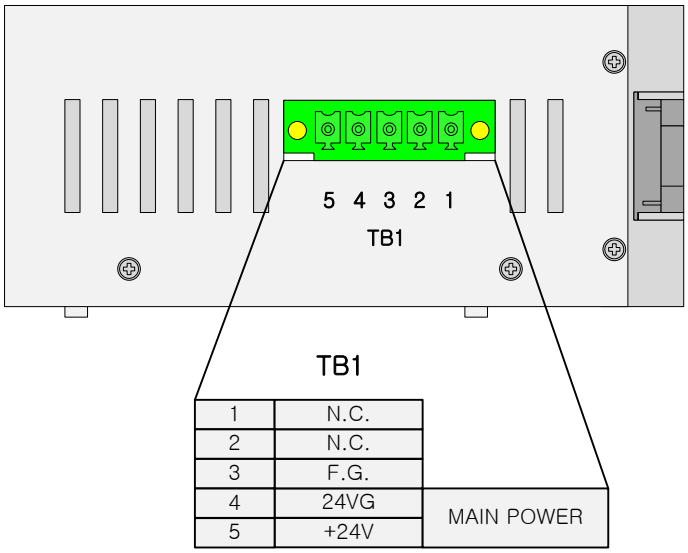


Fig. 27. Pin Array of Bottom Panel Power Connector of RTEX-AI8 Module (TB1)

As shown in Fig. 27, there is a power input connector (TB1) at the bottom panel of RTEX- AI8. One five pin MC 1,3/5-GF-3,5THT connector of Phoenix is used for power input connector. Input power is 24V DC. Reverse polarity protection diode is installed inside the module so that power is not supplied to the module and the module will not function if polarity is reversed. Pin 1 and 2 are not connected and pin 3, 4 and 5 are for external power.

4.5. RTEX-AO4 Module(To be released)

4.5.1. Pin Array and Description of Front Panel Connector (ACN1)

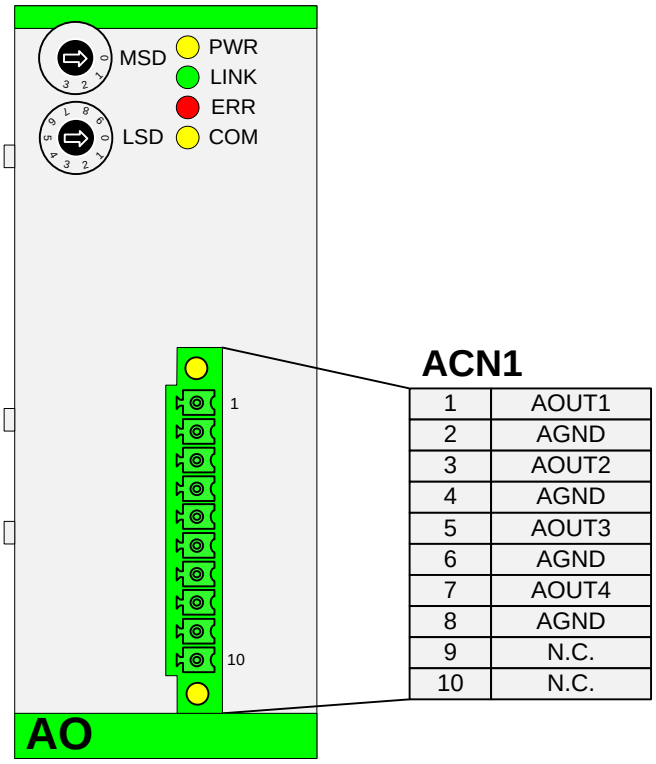


Fig. 28. RTEX-AO4 Front Panel Connector (ACN1) Pin Array

As shown in Fig. 28, RTEX-A04 is an analog output function module that produces four channels of -10V ~ +10V analog output signal and the pins are arranged in such a way that pin 1 and pin 2 are for channel 1, pin 3 and pin 4 are for channel 2, pin 5 and pin 6 are for channel 3 and pin 7 and pin 8 are for channel 4. The 10 pin MCD 1,5/16-GIF-3,81 connector of Phoenix is used for ACN1 with the pin numbering staring from top as pin 1 and bottom as pin 10. Refer to **"5.5. Major Functions of RTEX-AO4 Module"** for detailed internal circuit.

4.5.2. Pin Array and Description of Front Panel LED Display

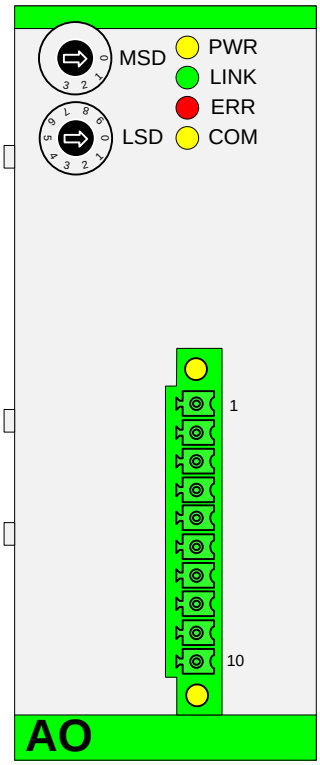


Fig. 29. RTEX-AO4 Front Panel LED Display

As shown in Fig. 29, there are four LED at the upper part and 32 LED on the right side of the front panel of RTEX-AO4 Module. Four LED at the upper part display the status of the slave module and the meaning of each light is as followings.

1) PWR (POWER) : It is on when the power is supplied to the module.
2) LINK (LINK) : It is on when the network is connected.
3) ERR (ERROR) : It displays error status.
4) COM (COMMUNICATION) : It displays the status of RTEX data communication.

Refer to **"5.5.1. Configuration and Description of Analyg Input Circuit of RTEX-AO4"** for detailed internal circuit.

4.5.3. Description of Top Panel MODULE ID

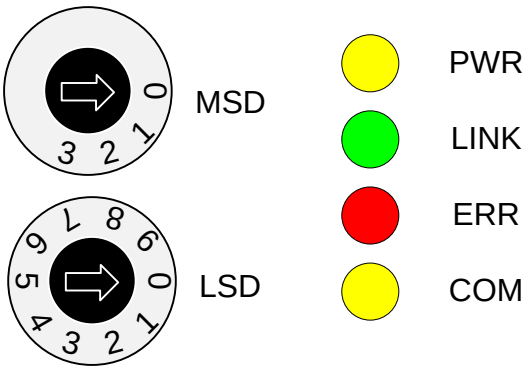


Fig. 30. . MODULE ID Switch of RTEX-AO4 Module

As shown in Fig. 30, there are switches to set the Module ID at the top panel of the RTEX-A04 Module. Up to 32 slave modules can be connected to one master board and the module ID should be set to 0 to 31. One of the most important things in setting node ID is to set a unique node ID that is not overlapped with other function modules. If there are duplicated ID within a node, it causes an error and entire modules may not work or system may not function.

4.5.4. Pin Array and Description of Bottom Panel Connector (TB1)

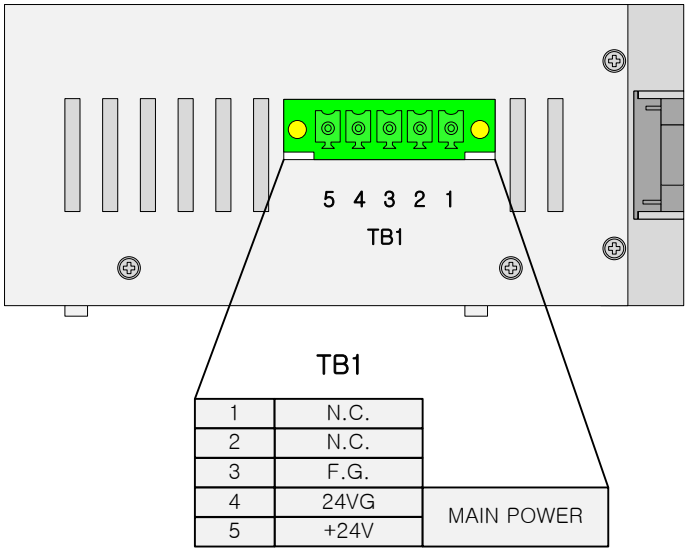


Fig. 31. Pin Array of Bottom Panel Power Connector of RTEX-AO4 Module (TB1)

As shown in Fig. 31, there is a power input connector (TB1) at the bottom panel of RTEX-AO4. One five pin MC 1,3/5-GF-3,5THT connector of Phoenix is used for power input connector. Input power is 24V DC. Reverse polarity protection diode is installed inside the module so that power is not supplied to the module and the module will not function if polarity is reversed. Pin 1 and 2 are NC and pin 3, 4 and 5 are for external power. Pin 1 and 2 are for IO power and pin 4 and 5 are for external power of the module.

5. Functional Description of RTEX Series Modules

5.1. Major Functions of RTEX-PM Module

5.1.1. Configuration and Description of Pulse Output Circuit

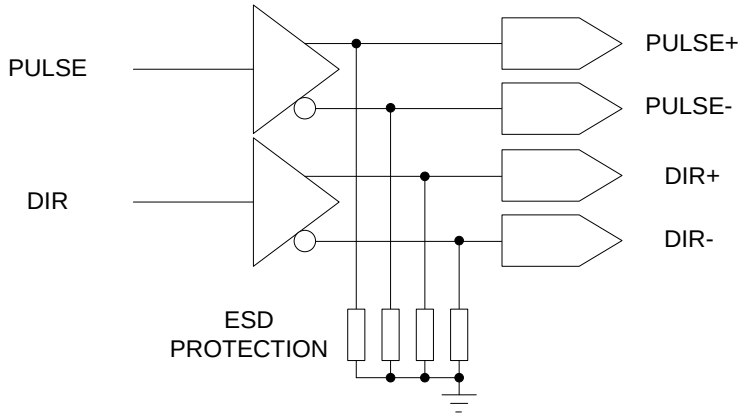


Fig. 32. Configuration of Pulse Output Circuit of RTEX-PM Module

Line transmitter circuit is included internally as shown in Fig. 32 and each signal line includes protection circuit to protect the devices from damage by ESD (Electric Shock Discharge) including static electricity. The voltage level of these signals is 5VDC.

5.1.2. Configuration and Description of Encoder Input Circuit

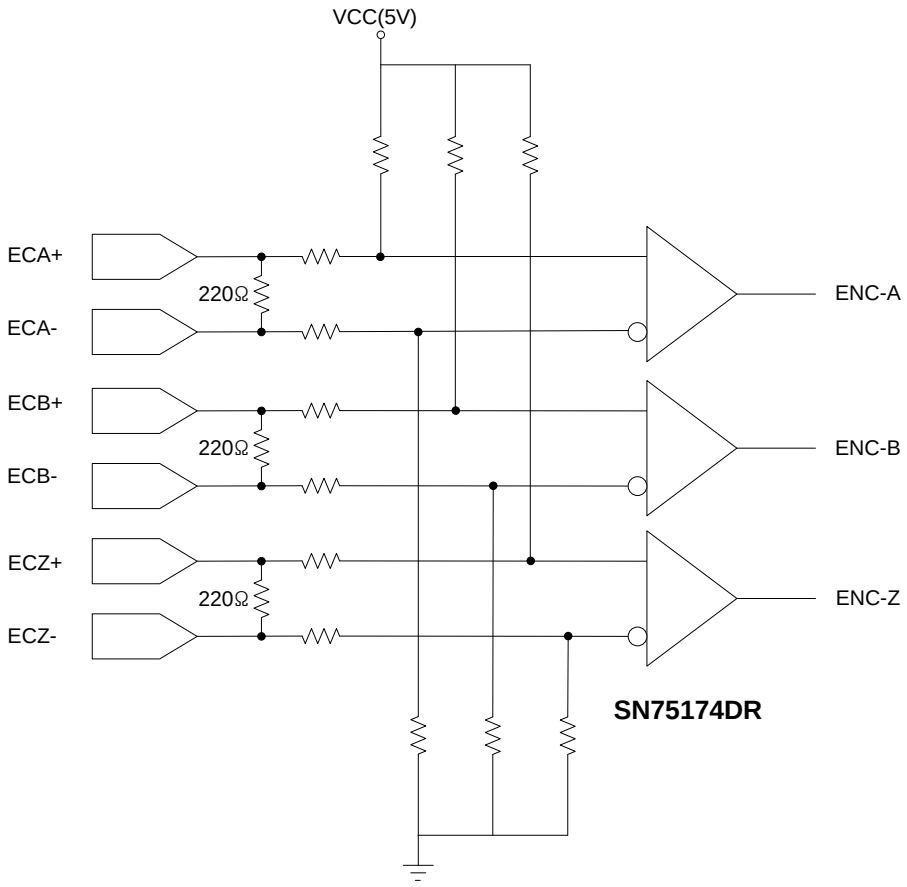


Fig. 33. Configuration of Encoder Input Circuit of RTEX-PM Module

As shown in Fig. 33, signal from encoder is received through line receiver circuit using 75174. The voltage level of these signals is 5VDC and up to 5MPPS input can be handled.

5.1.3. Configuration and Description of Motion Digital Input and Output Circuit

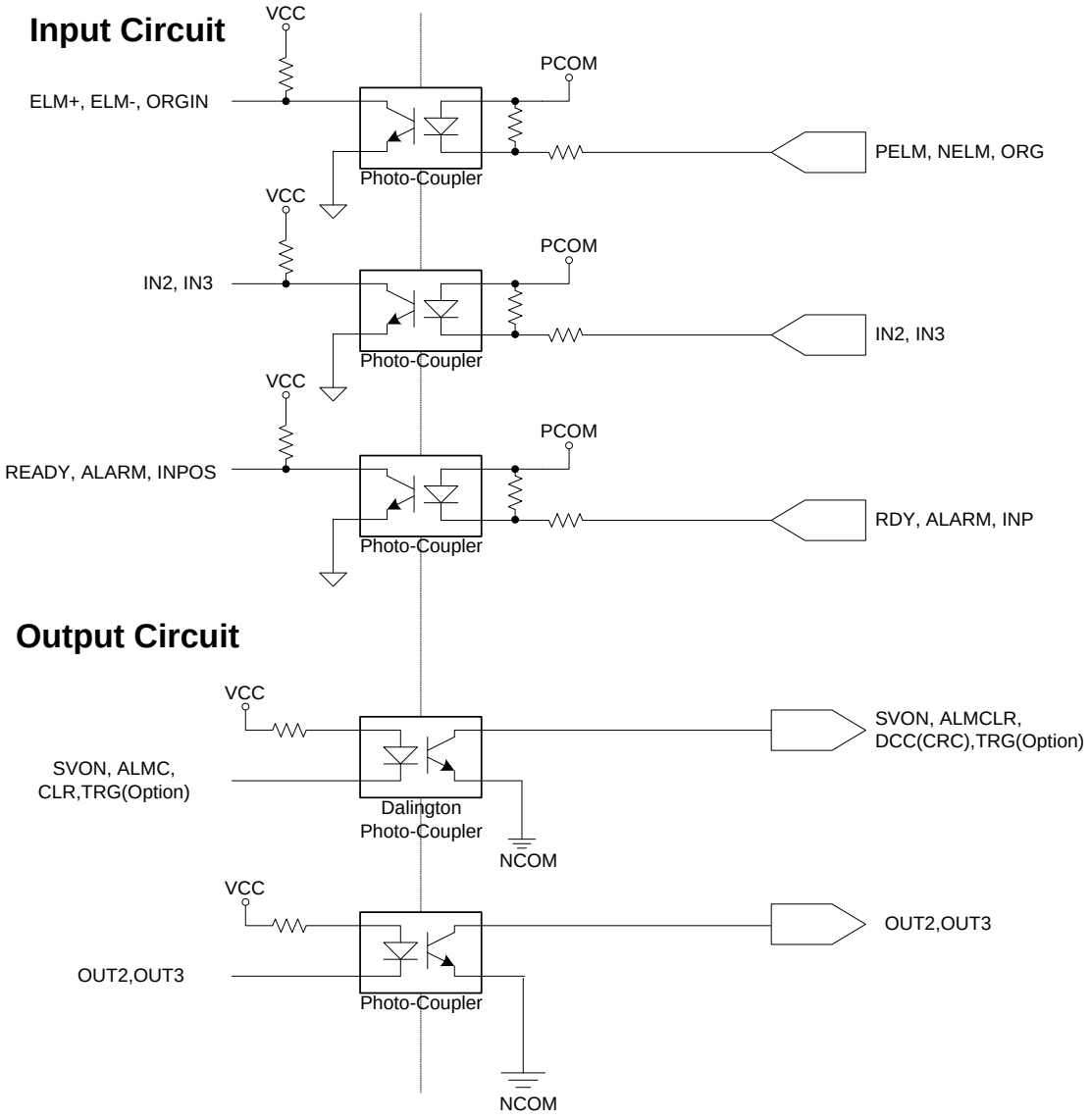


Fig. 34. Configuration if Motion Digital Input and Output Circuit of RTEX-PM Module

RTEX-PM Module includes internal motion digital input and output circuit as shown in Fig 34. Input circuit is configured in the same way as the universal input circuit. Output circuit is configured using driver IC for universal outputs and Darlington photo-coupler is used for other signals including SERVO ON(SVON), ALARM CLEAR(ALMCLR), DEFFEENTAIL COUNTER CLEAR(DCC, CRC) and TRIGGER(TRG).

5.1.4. Configuration and Description of MPG Input Circuit

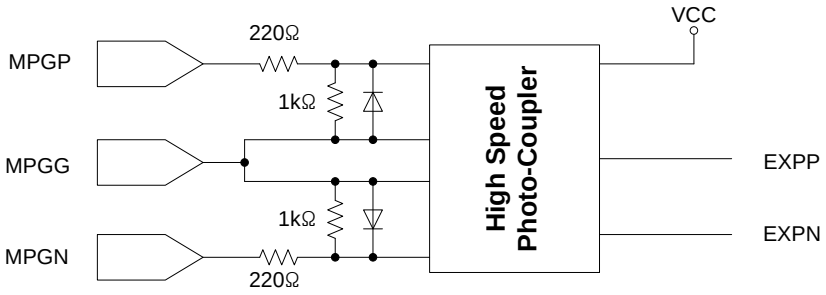


Fig. 35. Configuration of MPG Input Circuit of RTEX-PM Module

RTEX-PM Module includes internal MPG input circuit as shown in Fig 35 and the input circuit is configured in the same way as the encoder input circuit. Connect a external current limiting resistor (Example: 300Ω for 12V, : 1kΩ for 24V) to the output signal when using 12V or 24V MPG.

5.1.5. Configuration and Description of TRIGGER Output Circuit

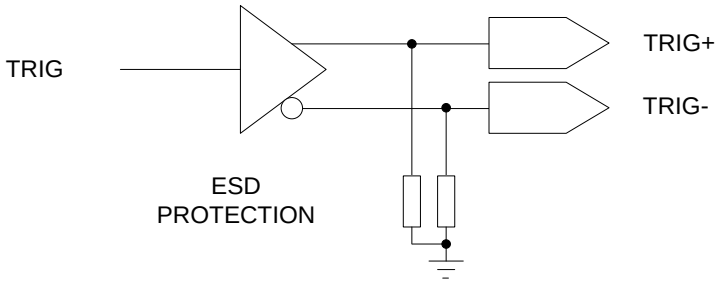


Fig. 36. Configuration of TRIGGER Output Circuit of RTEX-PM Module

Line transmitter circuit is included internally as shown in Fig. 36 and each signal line includes protection circuit to protect the devices from damage by ESD (Electric Shock Discharge) including static electricity. The voltage level of these signals is 5VDC.

5.1.6. Configuration and Description of Emergency Stop Circuit

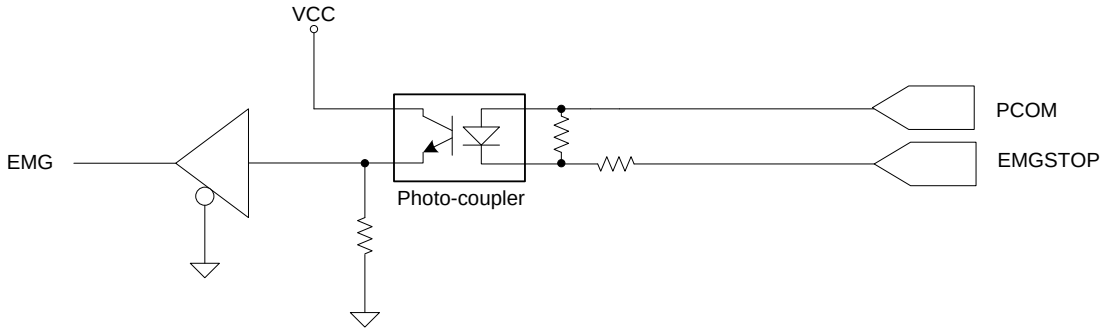


Fig. 37. Configuration of Emergency Stop Circuit of RTEX-PM Module

Configure the emergency stop circuit so that there is at least 2mA current flow at the input stage when using emergency stop input for the RTEX-PM module as shown in Fig 37. Input with the current flow less than that may not be accepted as valid input. ***Since PCOM used for EMG STOP is not connected to PCOM (DCN1 Connector) of I/O Power Supply in hardware, make sure to connect it to PCOM for motion control during wiring or consider it for other ways of application.***

Table 6 and Table 7 show pin array of MCN1 and MDCN1 connectors of RTEX, and Fig 37 shows MCN1, MCN2 and MDCN12, motion input and output connectors of RTEX modules.

Table 6. Pin Array of RTEX-PM MCN1 Connector

Pin Number	Signal Name	Description of signal	Pin Number	Signal Name	Description of Signal
1	**PCOM**	**POSITIVE COMMON**	14	N.C.	NO CONNECTION
2	INPOS	INPOSITION	15	N.C.	NO CONNECTION
3	SVON	SERVO ON	16	N.C.	NO CONNECTION
4	ALARM	ALARM	17	N.C.	NO CONNECTION
5	ALMCLR	ALARM CLEAR	18	N.C.	NO CONNECTION
6	RDY(IN4)	READY	19	ENC-A+	ENCODER A+ PHASE
7	DCC(CRC)	DEFFERNTIAL COUNTER CLEAR	20	ENC-A-	ENCODER A- PHASE
8	BRK(OUT4)	BRAKE	21	ENC-B+	ENCODER B+ PHASE
9	**NCOM**	**NEGATIVE COMMON**	22	ENC-B-	ENCODER B- PHASE
10	**LGND**	**LOGIC GROUND**	23	ENC-Z+	ENCODER Z+ PHASE
11	**ENCPWR**	**ENCODER POWER(5V)**	24	ENC-Z-	ENCODER Z- PHASE
12	PULSE+	POSITIVE PULSE	25	DIR+	POSITIVE DIRECTION

13	PULSE-	NEGATIVE PULSE	26	DIR-	NEGATIVE DIRECTION

Table 7. Pin Array of RTEX-PM MDCN1

Pin Number	Signal Name	Description of signal	Pin Number	Signal Name	Description of Signal
1	x_MPGP	MPG A PHASE	14	N.C	NO CONNECTION
2	x_MPGN	MPG B PHASE	15	N.C	NO CONNECTION
3	x_MPGG	MPG COMMON GROUND	16	N.C	NO CONNECTION
4	x_PELM	POSITIVE END LIMIT	17	N.C	NO CONNECTION
5	x_NELM	NEGATIVE END LIMIT	18	N.C	NO CONNECTION
6	x_ORG	ORIGIN	19	N.C	NO CONNECTION
7	x_IN2	GENERAL INPUT 2	20	N.C	NO CONNECTION
8	x_IN3	GENERAL INPUT 3	21	N.C	NO CONNECTION
9	x_OUT2	GENERAL OUTPUT 2	22	N.C	NO CONNECTION
10	x_OUT3	GENERAL OUTPUT 3	23	N.C	NO CONNECTION
11	x_TRG+	DIFFERENTIAL POSITIVE TRIGGER	24	N.C	NO CONNECTION
12	x_TRG-	DIFFERENTIAL NEGATIVE TRIGGER	25	N.C	NO CONNECTION
13	**PCOM**	**POSITIVE COMMON**	**26**	**NCOM**	**NEGATIVE COMMON**

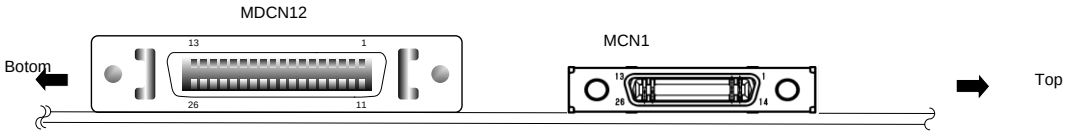


Fig. 38. Motion Input / Output Connector MCN1, MDCN1 of RTEX-PM Module

5.2. Major Functions of RTEX-DI32 Modules

5.2.1. Configuration and Description of Digital Input Circuit

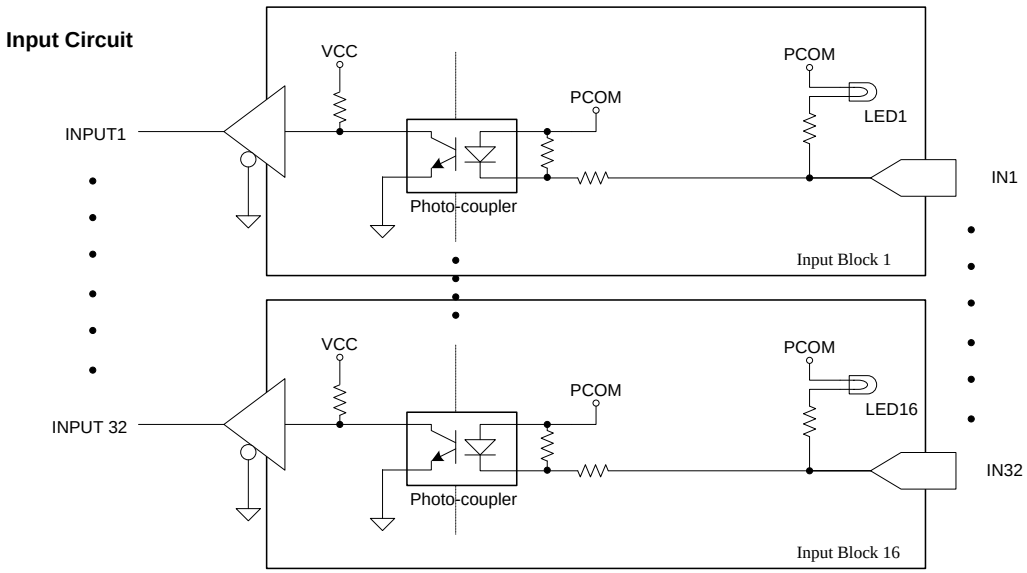


Fig. 39. Configuration of Internal Input Circuit of RTEX-DI32 Module

RTEX-DI32 Module includes 32 of digital input circuits as shown in Fig 39. The module is electrically isolated from outside through photo-couplers.

Configure the digital input circuit so that there is at least 2mA current flow at the input stage. Input with the current flow less than that may not be accepted as valid input.

Table 8 shows the pin array of the RTEX-DI32 DCN1 connector.

Table 8. Pin Array of RTEX-DI32 DCN1 Connector

Pin Number	Signal Name	Description of signal	Pin Number	Signal Name	Description of Signal
1	IN1	Input 1	17	IN17	Input 17
2	IN2	Input 2	18	IN18	Input 18
3	IN3	Input 3	19	IN19	Input 19
4	IN4	Input 4	20	IN20	Input 20
5	IN5	Input 5	21	IN21	Input 21
6	IN6	Input 6	22	IN22	Input 22
7	IN7	Input 7	23	IN23	Input 23
8	IN8	Input 8	24	IN24	Input 24
9	IN9	Input 9	25	IN25	Input 25
10	IN10	Input 10	26	IN26	Input 26
11	IN11	Input 11	27	IN27	Input 27
12	IN12	Input 12	28	IN28	Input 28
13	IN13	Input 13	29	IN29	Input 29
14	IN14	Input 14	30	IN30	Input 30
15	IN15	Input 15	31	IN31	Input 31
16	IN16	Input 16	32	IN32	Input 32

5.3. Major Functions of RTEX-DO32T Module

5.3.1. Configuration and Description of Digital Output Circuit

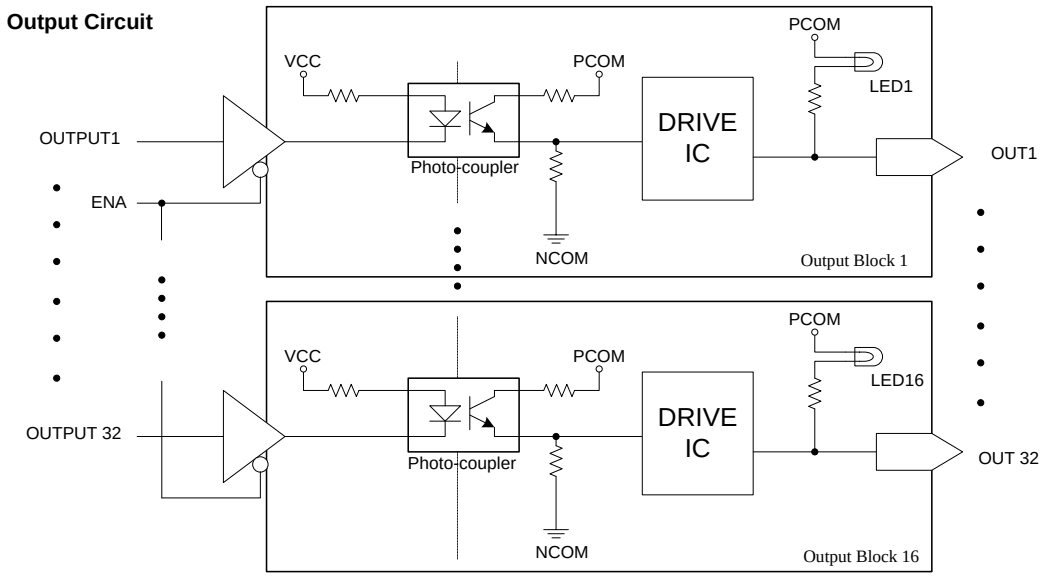


Fig. 40. Configuration of Internal Output Circuit of RTEX-DO32T Module

RTEX-DO32T Module includes 32 of digital output circuit blocks as shown in Fig 40. The module is electrically isolated from outside through photo-couplers.

Configure the digital output circuit to limit the current draw below 50 mA. If the current draw exceeds that, protection function of the drive IC will be activated and the operation of the connected sensor may stop.

Table 9 shows the pin array of the RTEX-DO32T DCN1 Connector.

Table 9. Pin Array of RTEX-DO32T DCN1 Connector

Pin Number	Signal Name	Description of signal	Pin Number	Signal Name	Description of Signal
1	OUT1	Output 1	17	OUT17	Output 17
2	OUT2	Output 2	18	OUT18	Output 18
3	OUT3	Output 3	19	OUT19	Output 19
4	OUT4	Output 4	20	OUT20	Output 20
5	OUT5	Output 5	21	OUT21	Output 21
6	OUT6	Output 6	22	OUT22	Output 22
7	OUT7	Output 7	23	OUT23	Output 23

8	OUT8	Output 8	24	OUT24	Output 24
9	OUT9	Output 9	25	OUT25	Output 25
10	OUT10	Output 10	26	OUT26	Output 26
11	OUT11	Output 11	27	OUT27	Output 27
12	OUT12	Output 12	28	OUT28	Output 28
13	OUT13	Output 13	29	OUT29	Output 29
14	OUT14	Output 14	30	OUT30	Output 30
15	OUT15	Output 15	31	OUT31	Output 31
16	OUT16	Output 16	32	OUT32	Output 32

5.4. Major Functions of RTEX-AI8Module (To be released)

5.4.1. ANALOG Configuration and Description of Input Circuit

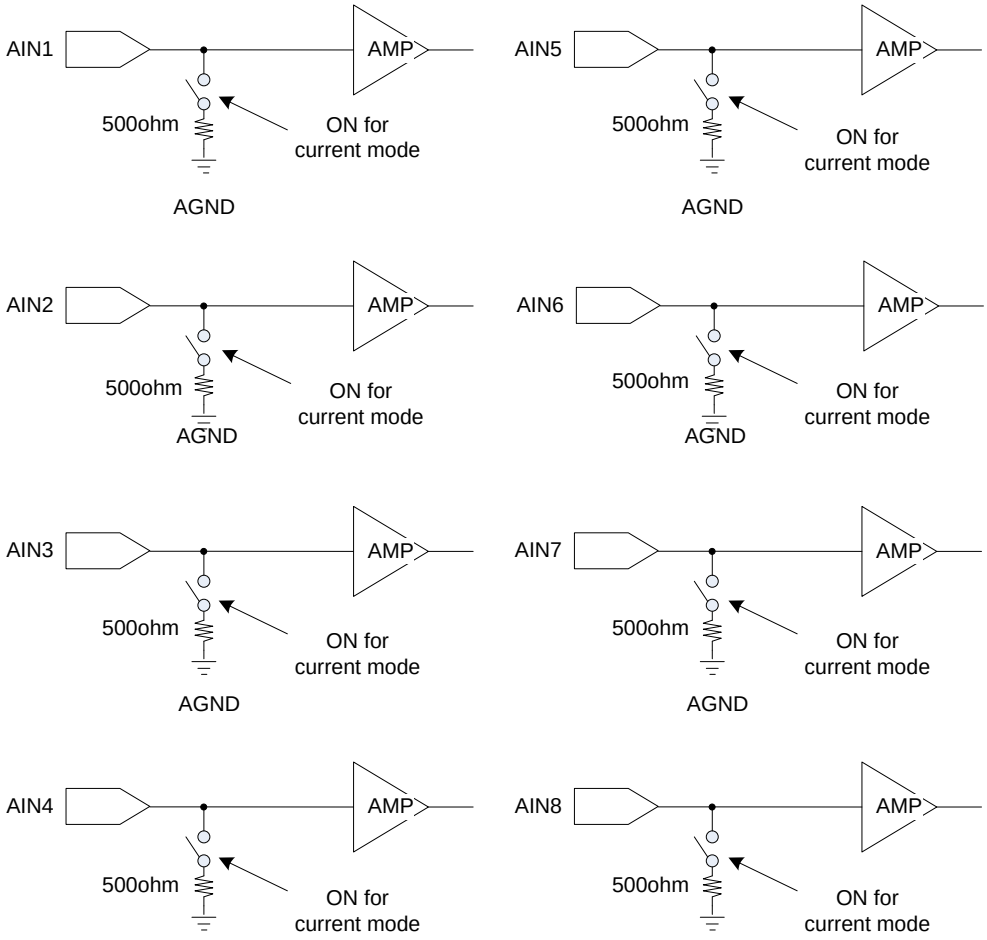


Fig. 41. Configuration of Internal Analog Input Circuit of RTEX-AI8 Module

RTEX-AI8 Module includes eight of analog input circuits as shown in Fig 41. Each circuit has a switch to select input mode for selecting either voltage mode or current mode. The operation will be voltage mode if the switch is set to OFF and current mode if it is set to ON.

Range of input can be programmed to one of -10V ~ +10V , 0 ~ +5V and 0 ~ +10V when the voltage mode is selected and the 4~20 mA current signal can be received by setting the DIP switch located on the front side to ON.

Table 10 shows the pin array of the ACN1 connector of RTEX-AI8 Module.

Table 10. Pin Array of ACN1 Connector of EIP-AI8 Module

Pin Number	Signal Name	Description of signal	Pin Number	Signal Name	Description of Signal
1	AIN1	Analog Input 1	11	AIN5	Analog Input 5
2	AGND	Analog Ground	12	AGND	Analog Ground
3	AIN2	Analog Input 2	13	AIN6	Analog Input 6
4	AGND	Analog Ground	14	AGND	Analog Ground
5	AIN3	Analog Input 3	15	AIN7	Analog Input 7
6	AGND	Analog Ground	16	AGND	Analog Ground
7	AIN4	Analog Input 4	17	AIN8	Analog Input 8
8	AGND	Analog Ground	18	AGND	Analog Ground
9	N.C.	NO CONNECTION	19	N.C.	NO CONNECTION
10	N.C.	NO CONNECTION	20	N.C.	NO CONNECTION

5.5. Major Functions of RTEX-AO4 Module (To be released)

5.5.1. Configuration and Description of Analog Output Circuit of RTEX-AO4 Module

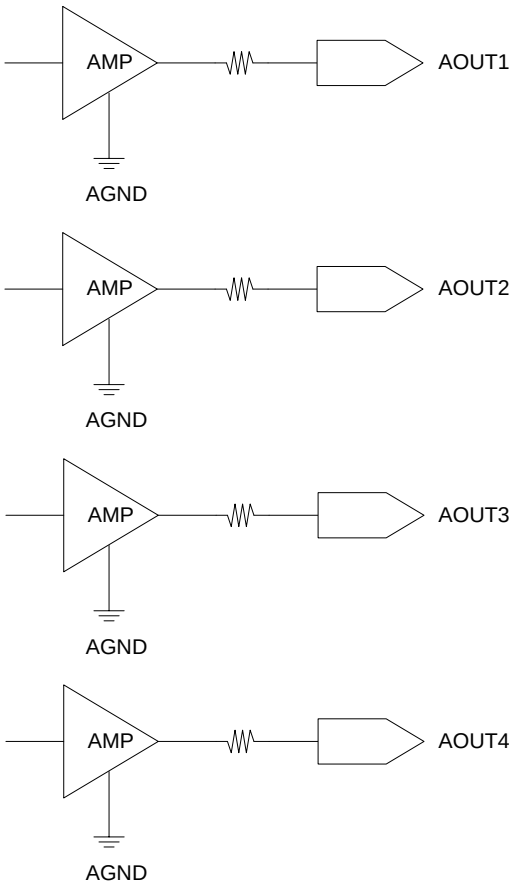


Fig. 43. Configuration of Internal Analog Output Circuit of RTEX-AO4 Module

RTEX-AO4 Module includes four of analog output circuits as shown in Fig 43. Each circuit has output protection resistor.

Table 11 shows the pin array of AC1 connector of RTEX-AO4 Module.

Table 11. Pin Array of ACN1 Connector of EIP-AO4 Module

Pin Number	Signal Name	Description of signal
1	AOUT1	Analog Output 1
2	AGND	Analog Ground
3	AOUT2	Analog Output 2
4	AGND	Analog Ground
5	AOUT3	Analog Output 3
6	AGND	Analog Ground
7	AOUT4	Analog Output 4
8	AGND	Analog Ground
9	N.C.	NO CONNECTION
10	N.C.	NO CONNECTION

6. Examples of RTEX Series Module Connection

6.1. Examples of RTEX-PM MCN1 Connection

6.1.1. Example of MITSUBISHI MR-J2-xxA Connection

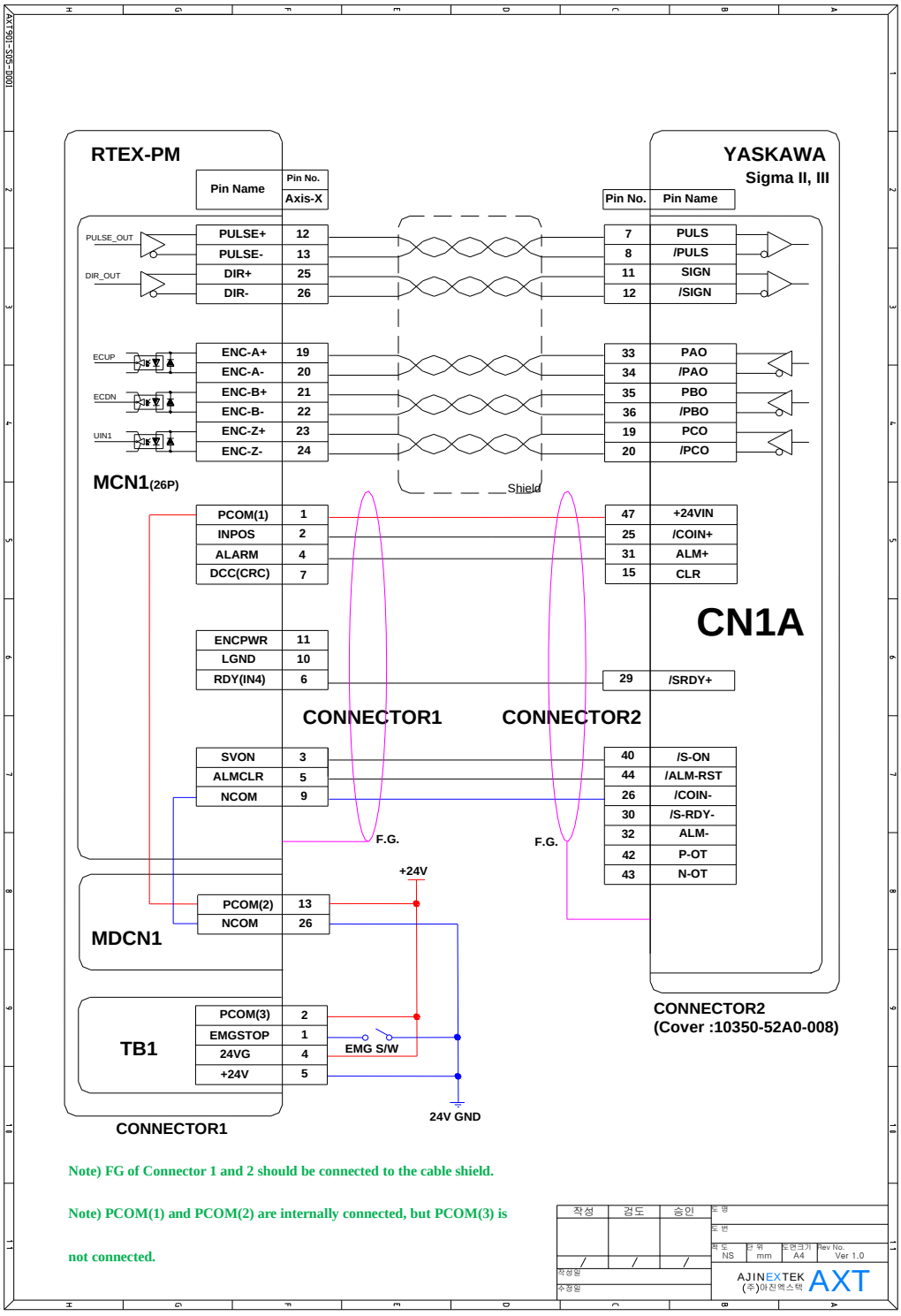


Fig. 44. Example Connection Between RTEX-PM and MITSUBISHI MR-J2-xxA

6.1.2. Example of MITSUBISHI MR-J3-xxA Connection

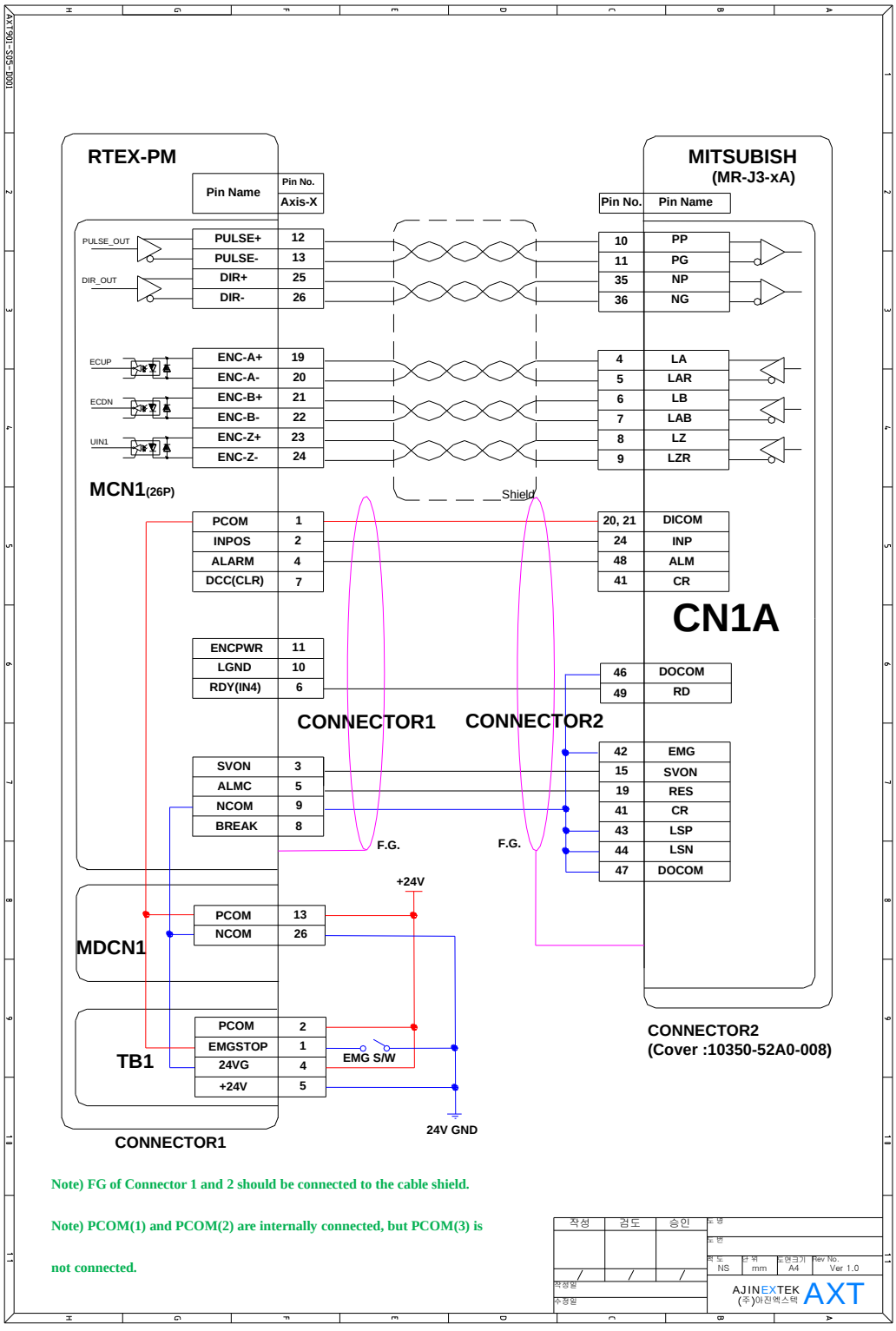


Fig. 45. Example Connection Between RTEX_PM and MITSUBISHI MR-J3-xxA

6.1.3. Example of PANASONIC MINAS A, A4 Series Connection

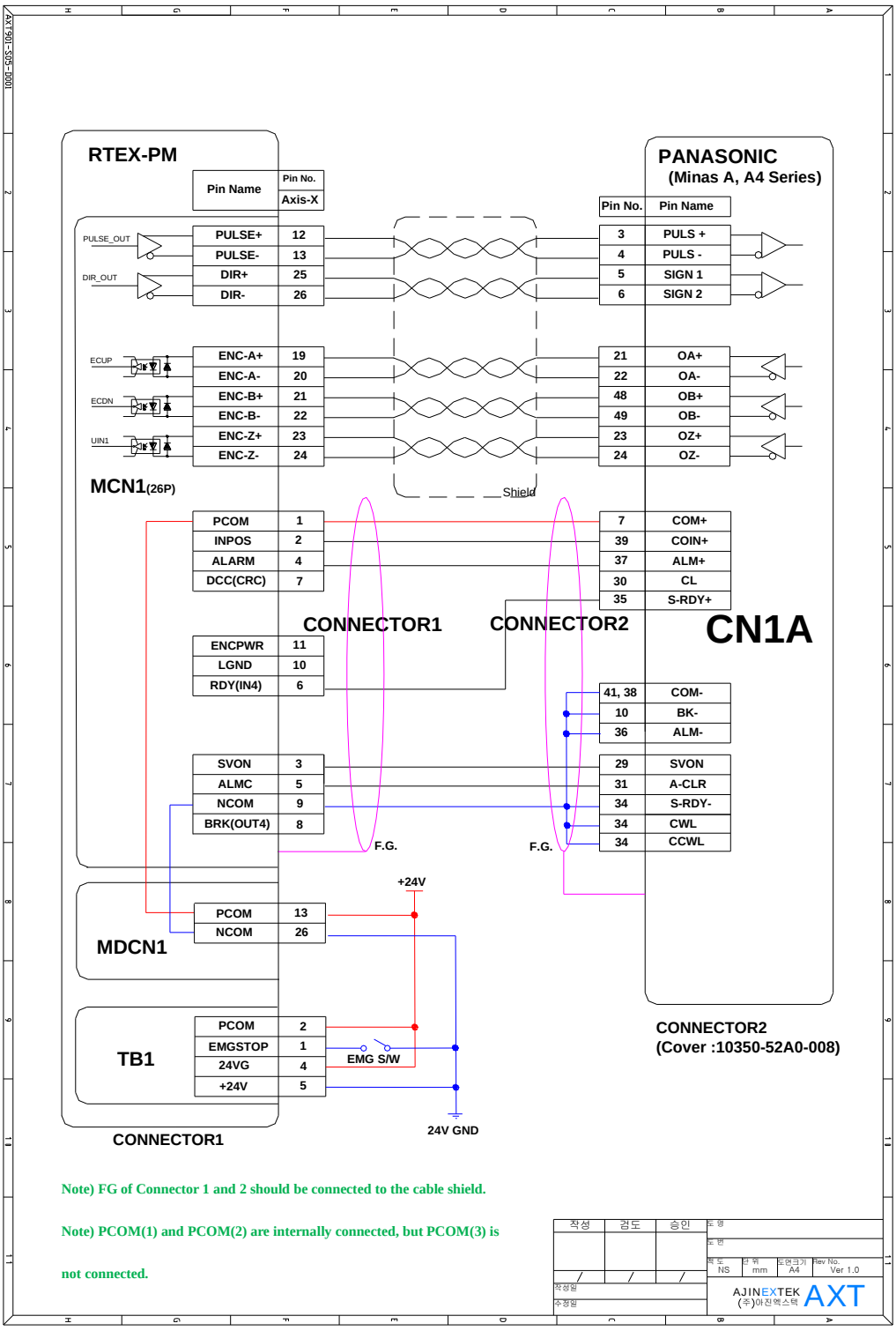


Fig. 46. Example Connection Between RTEX-PM and PANASONIC MINAS A, A4 Series

6.1.4. Example of YASKAWA SIGMA II, III Connection

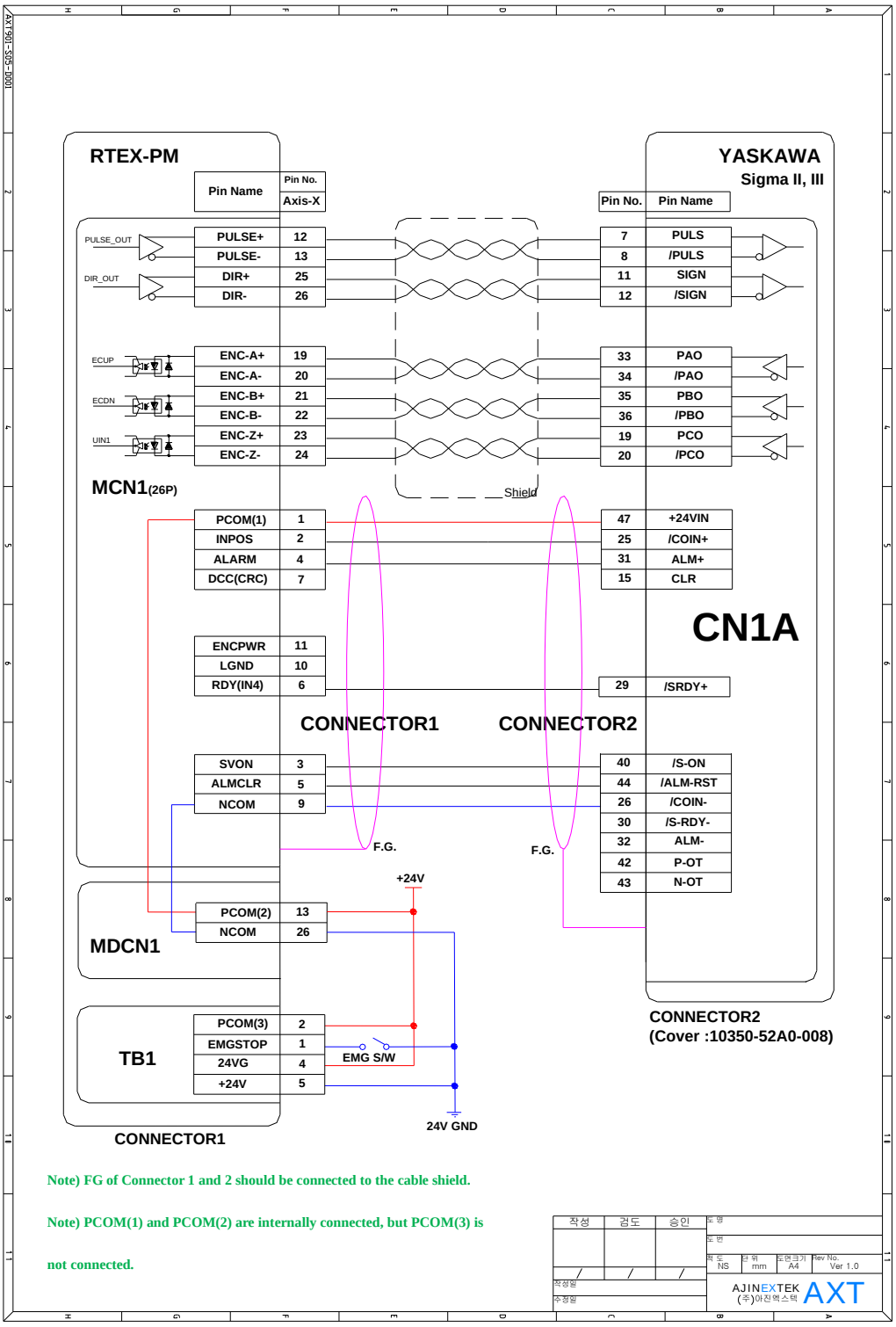


Fig. 47. Example Connection Between RTEX-PM and YASKAWA SIGMA II, III

6.1.5. Example of RTEX-PM IO MDCN1 Connection

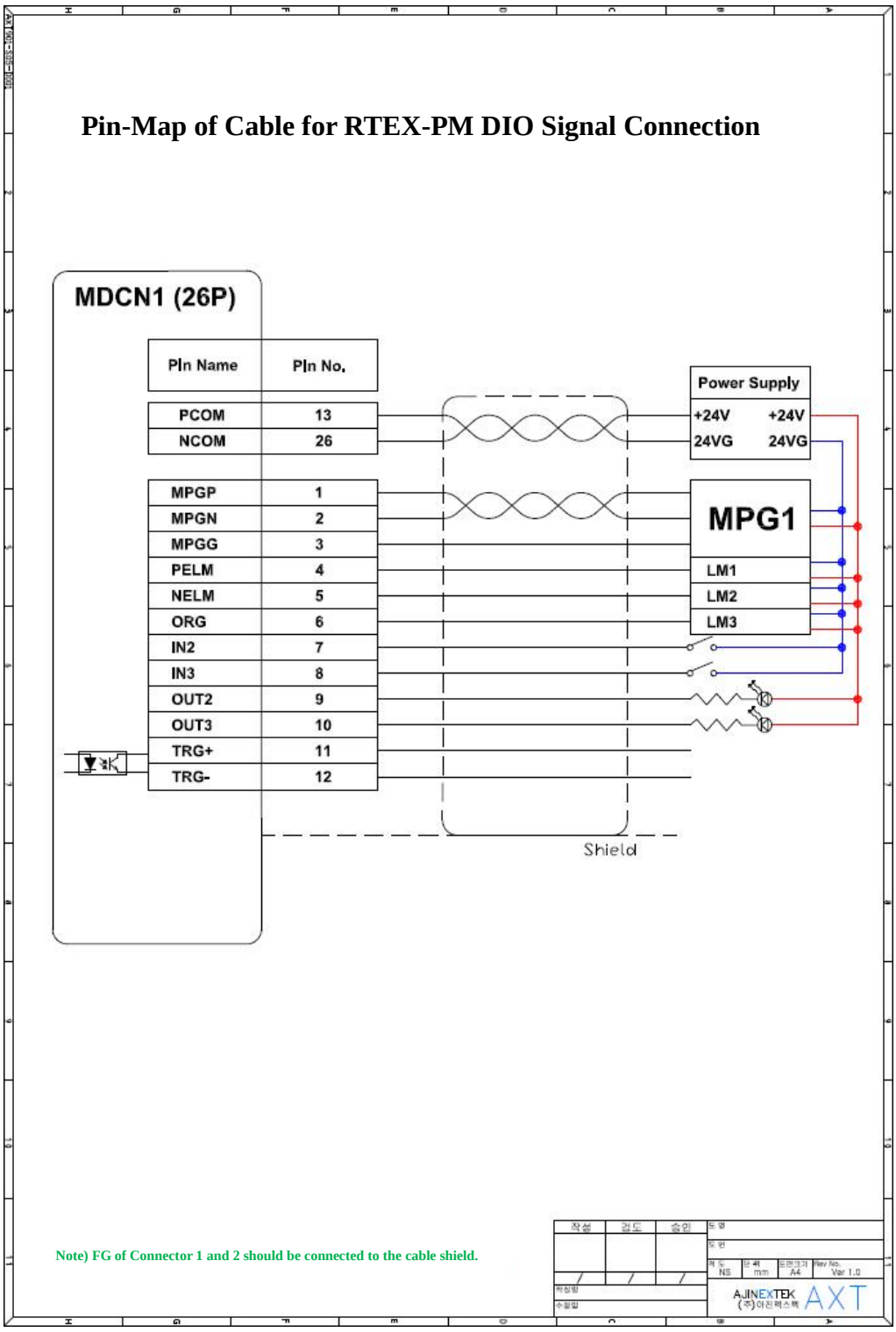


Fig. 48. Example of Digital Input and Output Connection of RTEX-PM MDCN1

6.1.6. Example of RTEX-DI32 DCN1 Connection

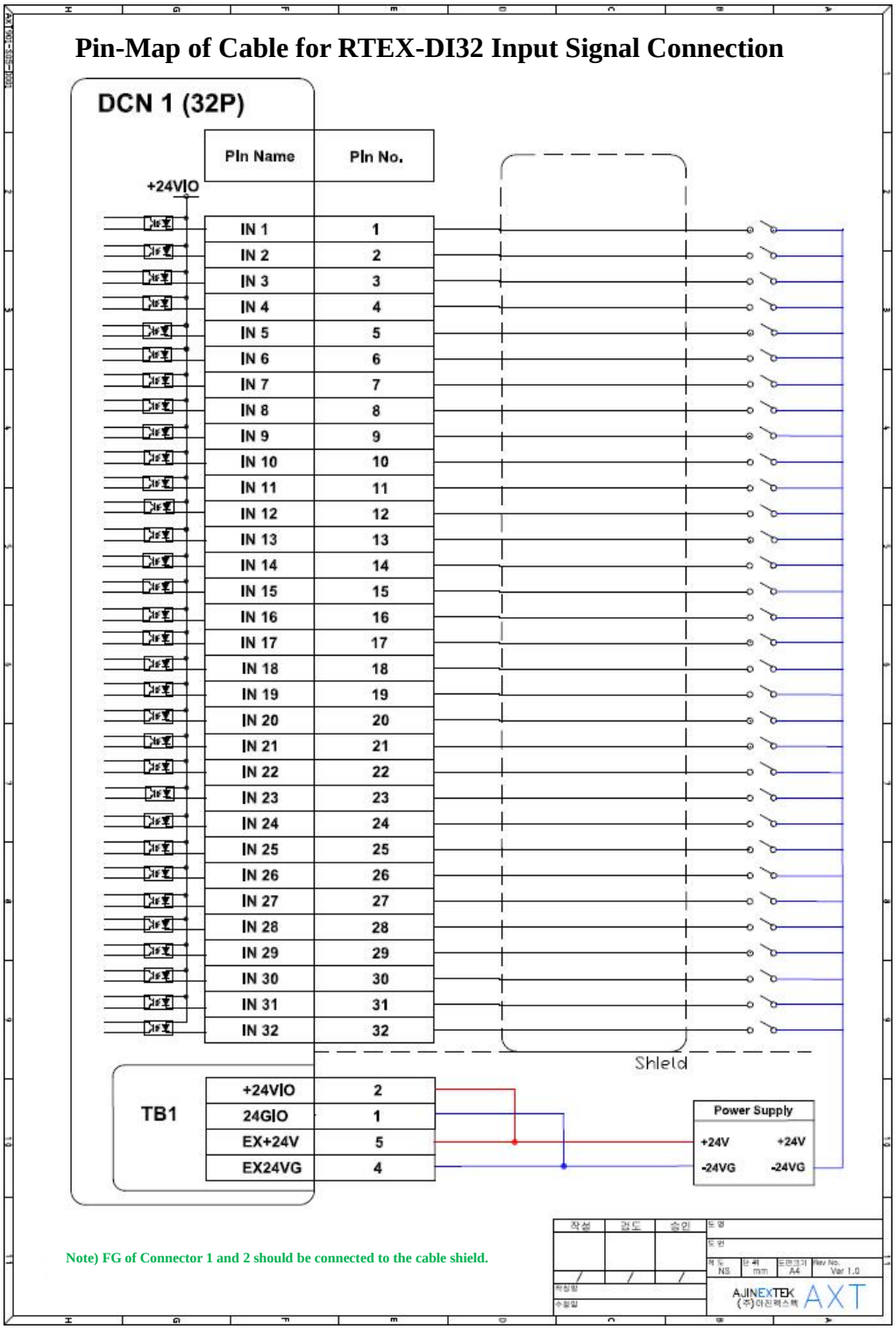


Fig. 49. Example of RTEX-DI32 Input Connection

6.1.7. Example of RTEX-DO32T DCN1 Connection

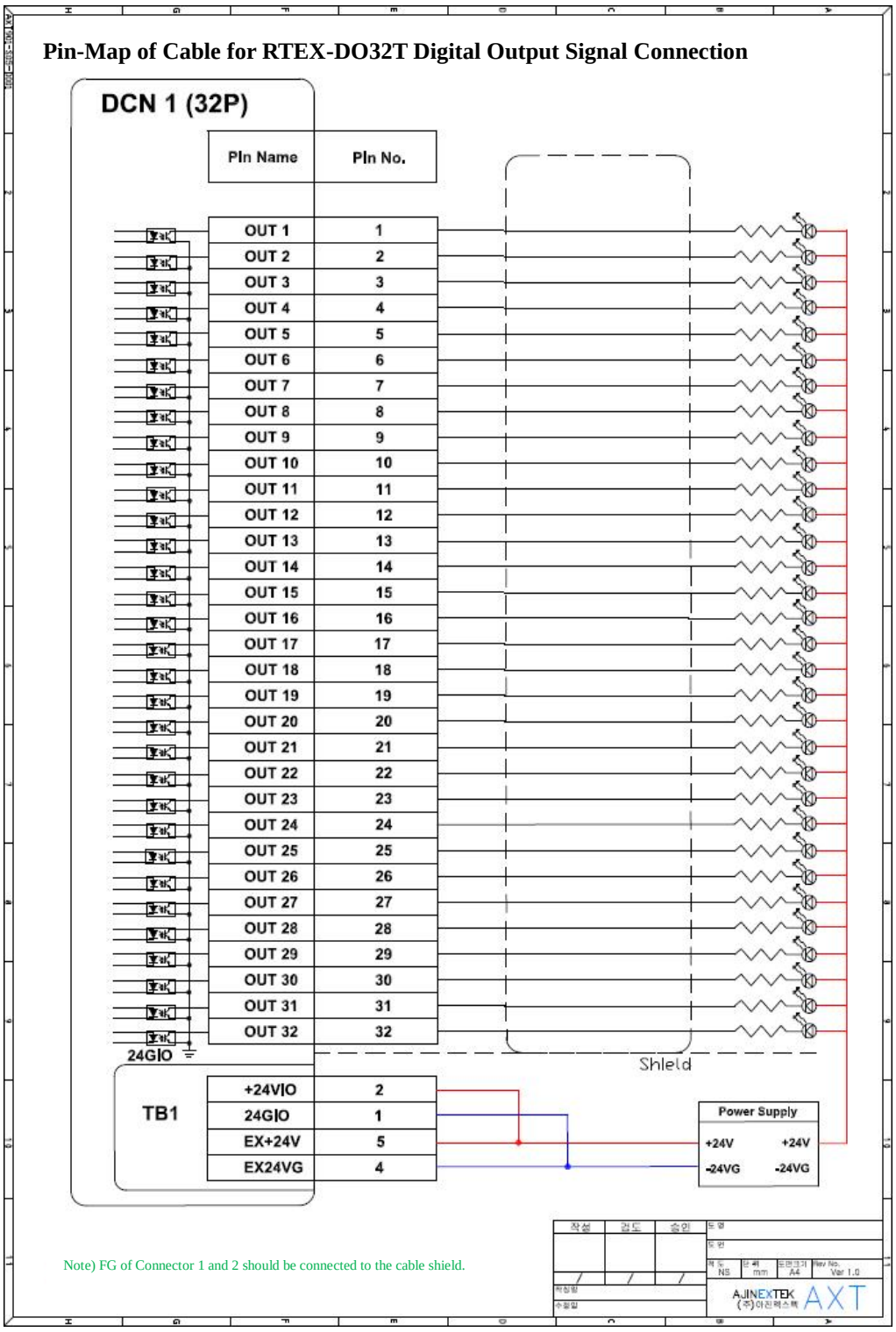


Fig. 50. Example of RTEX-DO32T Output Connection

6.1.8. Example of RTEX-AI8 (To be released) ACN1 Connection

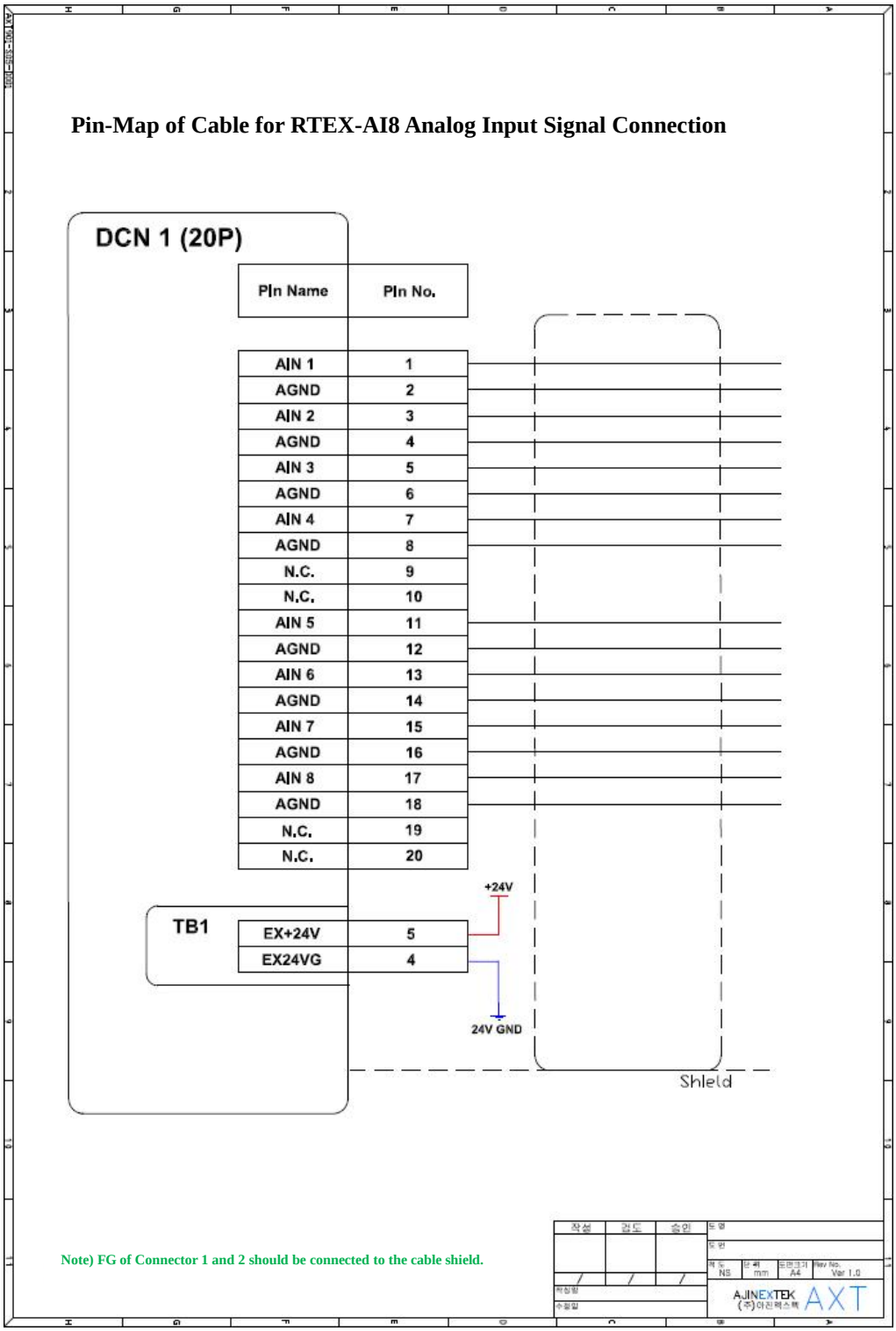


Fig. 51. Example of RTEX-AI8 Input Connection

6.1.9. Example of RTEX-AO (To be released) ACN1 Connection

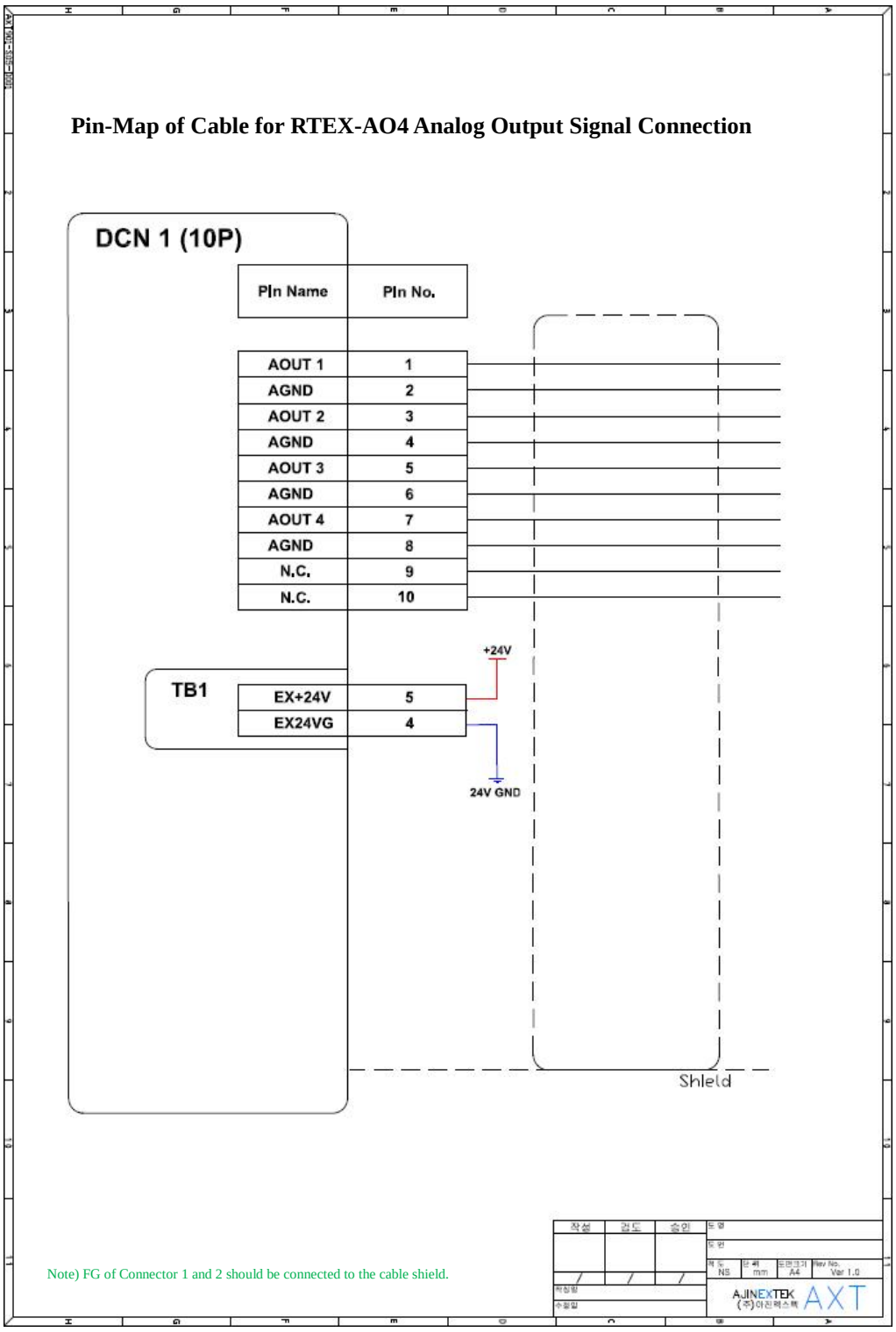


Fig. 52. Example of RTEX-AO4 Output Connection

7. Ordering Information (Family)

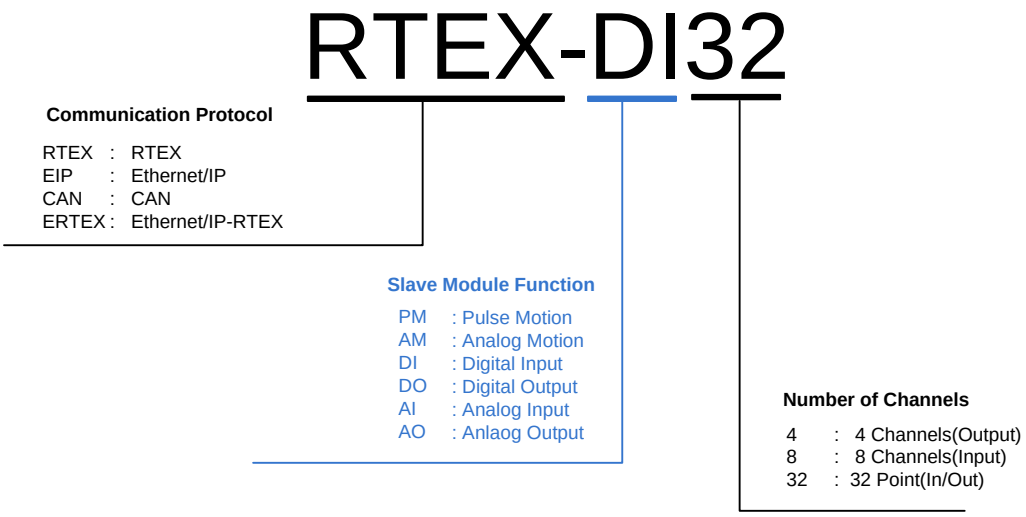


Fig. 53. How to read model names of RTEX Series

Table 12. Product Families of RTEX SEIRES Modules

Product Families	Description
RTEX-PM	RTEX PULSE MOTION MODULE (1 Axis)
RTEX-DI32	RTEX DIGITAL INPUT MODULE(DIGITAL INPUT 32 contacts)
RTEX-DO32T	RTEX DIGITAL OUTPUT MODULE(DIGITAL OUTPUT 32 contacts)
RTEX-AI8	RTEX ANALOG INPUT MODULE(8CH) **(To be released)**
RTEX-AO4	RTEX ANALOG OUTPUT MODULE(4CH) **(To be released)**
A4N	SERVO DRIVE of PANASONIC EXCLUSIVELY FOR RTEX

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